Supplementary information

External prognostic evaluation and myeloid state characterization of a five-gene score in acute myeloid leukemia

Cellular Oncology

Online Resource 1

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Online Resource 1 contains Supplementary Figs. S1–S13 with their legends and Supplementary Tables S1–S10. Processed analysis tables and the additional statistics cited below are in Online Resource 2 (ESM_2.zip); compact R objects are in Online Resource 3 (ESM_3.zip).

Candidate gene annotations

Supplementary Table S5 retains all 113 fixed candidates and original list memberships, together with gene-level evidence groups and reference identifiers. Category A indicates gene-specific original experimental support for the stated process; category B indicates pathway, AML background, or official functional annotation support; category C indicates insufficient corresponding support verified in the selected sources. Evidence-group descriptions define the precise scope of support. Category A does not automatically imply AML-specific evidence.

Literature annotations were added to the fixed lists on October 7, 2026. They were not used to reselect candidates and do not document how the lists were originally assembled. The 13 categories encompass cell death and related processes: autophagy-related mechanisms do not automatically establish autophagy-dependent death, and mitotic catastrophe is not an independent death modality. SLC16A4 corresponds to MCT5 and is interpreted in its transporter-family context. Complete references are provided in Supplementary Table S5c below.

Statistical units

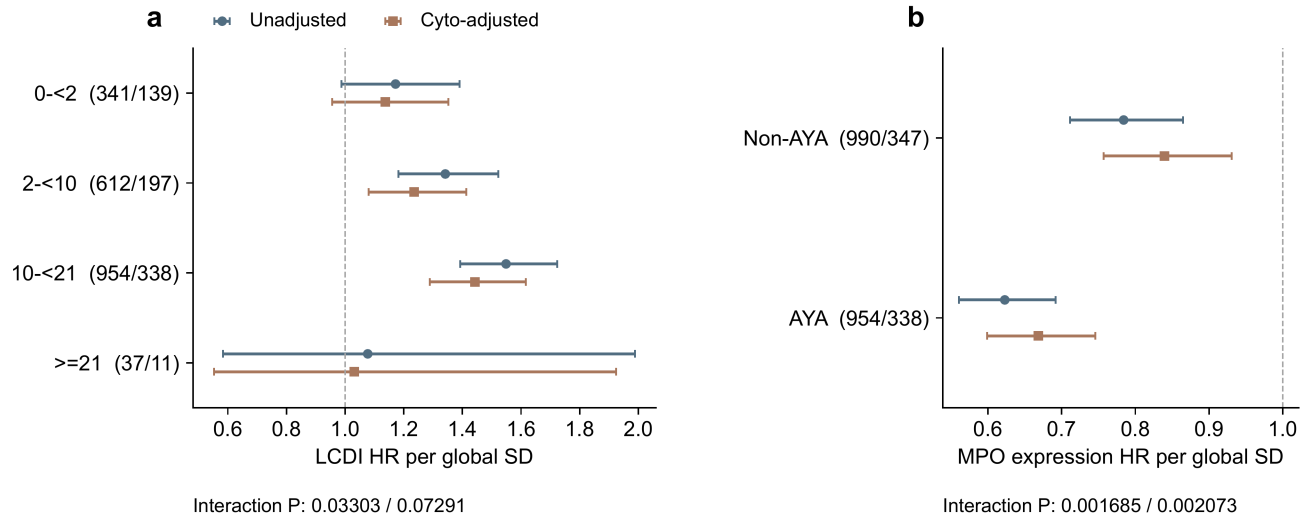
Patient-level inference and expression-record evaluation are reported separately. Supplementary Tables S1–S2 provide full-precision results for both LCDI and MPO-risk, with separate FDR columns for the eight-test family and 14-gene union. Supplementary Tables S6–S7 report paired clinical model increments in identical complete cases. Supplementary Figs. S7–S9 assess the non-MPO component conditional on MPO; these analyses use different baselines and are not interchangeable.

Supplementary Table S8 reports mean net benefit differences across the full 3-year DCA threshold range, intervals, raw P values, and FDRs. Meta-analysis in Supplementary Fig. S6 uses consistent Wald intervals; other survival figures and tables use intervals from their corresponding outputs. Resampling frequencies in Supplementary Figs. S10–S11 use 1,000 iterations per level as denominators. Supplementary Figs. S12–S13 aggregate Ct values by independent individual. Initial and remeasured-reference analyses share target-gene measurements and do not constitute independent validation cohorts.

Contents

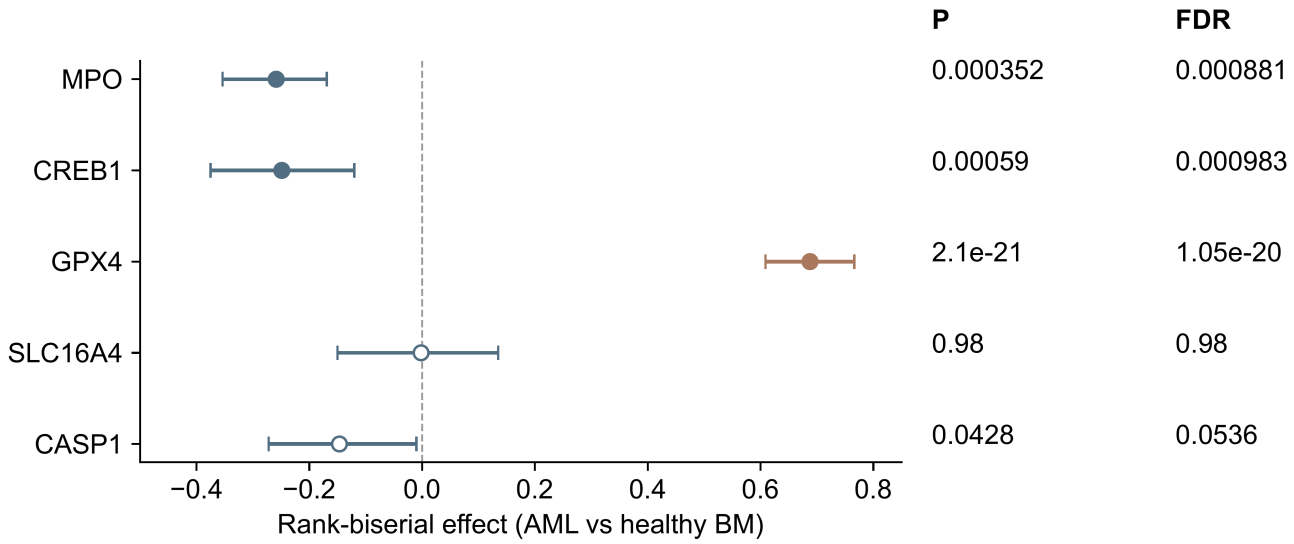
Figures S1–S13 appear first, followed by Tables S1–S10. Wide source tables are divided into consecutive column blocks with repeated source-row numbers and identifying fields; the blocks describe the same records and are not additional observations.

Supplementary Fig. S1



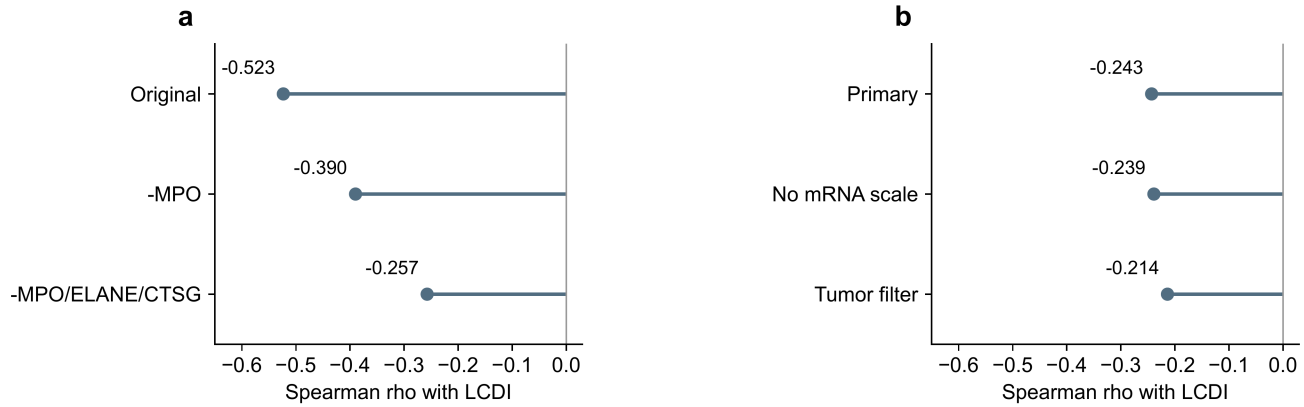
Age-related heterogeneity in TARGET and cytogenetic risk adjustment. a Stratum-specific HRs from LCDI Cox interaction models for ages 0–<2, 2–<10, 10–<21, and ≥21 years. b HRs for MPO expression in AYA (10–<21 years) and non-AYA groups. Parentheses show patients/deaths (total n=1,944; 685 deaths). Blue circles and orange squares denote unadjusted and cytogenetic risk-adjusted models; lines show 95% CIs. Both scores use the SD of the full TARGET cohort. MPO is oriented toward higher expression, not MPO-risk. Global LCDI interaction P/FDR values are 0.0330/0.0330 unadjusted and 0.0729/0.0729 adjusted; MPO values are 0.001685/0.003370 and 0.002073/0.004146. The two related interactions are BH-adjusted separately in each setting. Proportional hazards diagnostic P values for the LCDI interaction are approximately 0.046/0.047. Age heterogeneity remains exploratory, and this figure does not establish cross-cohort replication. Cyto, cytogenetic risk; SD, standard deviation

Supplementary Fig. S2



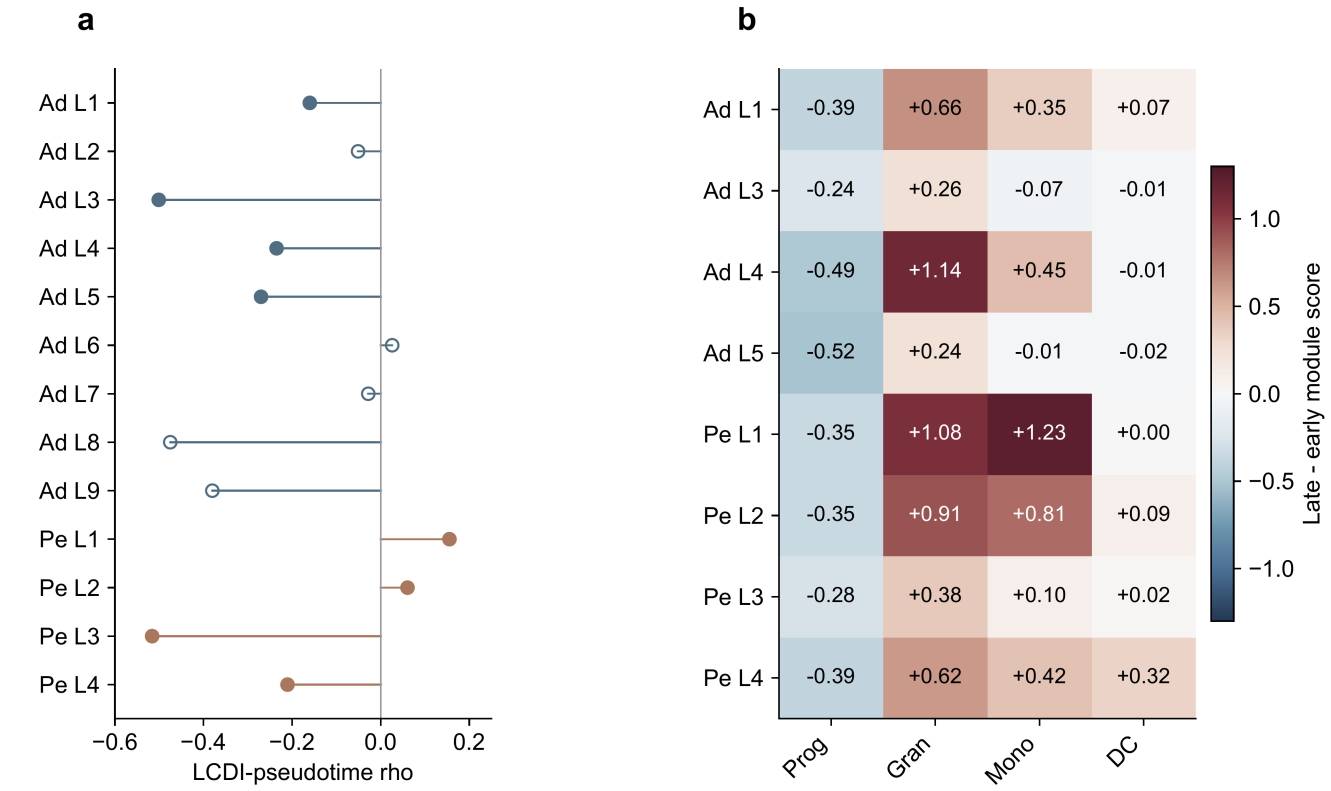
Five-gene expression differences in same-platform bone marrow data. GSE13159 includes 501 AML and 73 healthy bone marrow samples. For each gene, the specific _at probe with the highest mean signal across included samples was selected independently of group labels. Points show rank-biserial effect sizes, positive for higher AML expression and negative for lower expression; lines show 95% CIs from 2,000 patient bootstrap samples (seed=20260804). Two-sided Mann–Whitney P values and five-gene BH FDRs appear on the right; filled points indicate FDR<0.05. GPX4 is higher and MPO/CREB1 lower in AML. CASP1 provides a nominal signal (FDR=0.0536), whereas SLC16A4 is unsupported (FDR=0.9801). This comparison addresses AML–healthy expression directions, not prognostic coefficients, and differs from the TCGA–CD34+ cross-platform comparison. BM, bone marrow

Supplementary Fig. S3



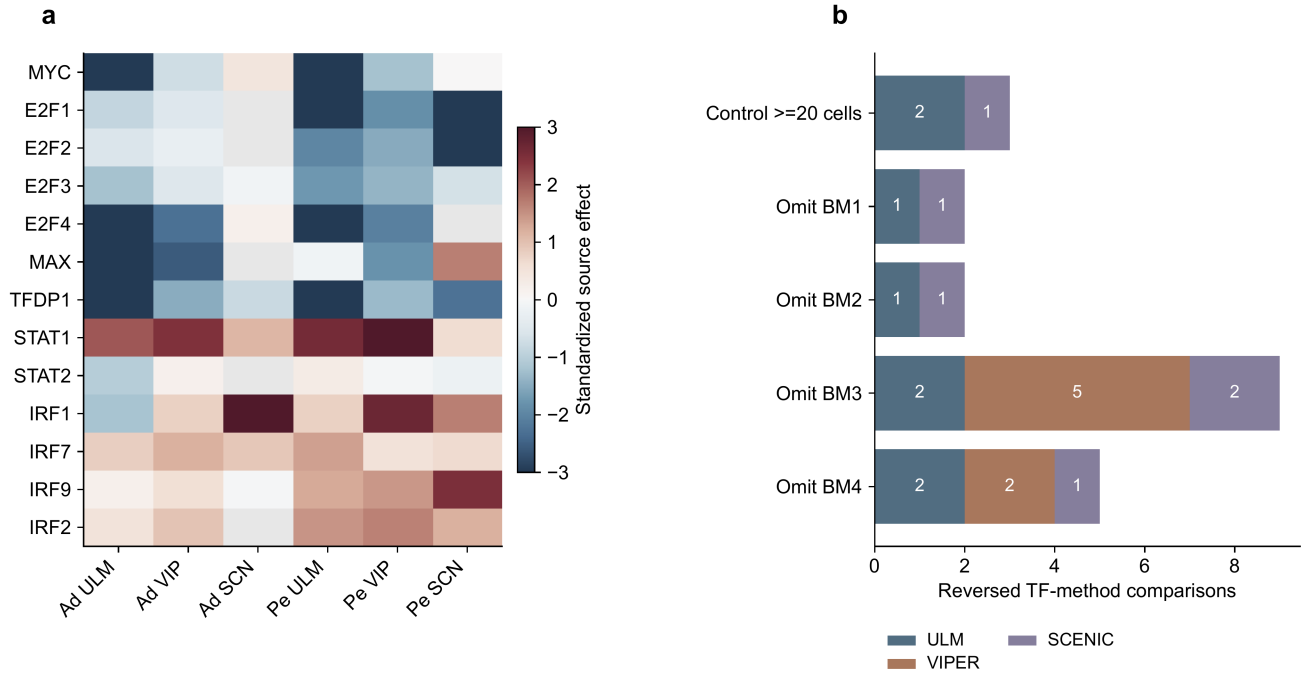
Overlap removal from myeloid signatures and sensitivity to deconvolution parameters. a Correlations of frozen LCDI with the full neutrophil signature, the signature excluding MPO, and the signature excluding MPO/ELANE/CTSG: rho=−0.5234, −0.3898, and −0.2573; FDR=4.21×10^{-41}, 8.69×10^{-22}, and 6.49×10^{-10}. b Correlations with the quantiSeq-inferred neutrophil component under primary settings, without mRNA scaling, and with tumor filtering: rho=−0.2430, −0.2388, and −0.2137; FDR=3.35×10^{-8}, 6.33×10^{-8}, and 2.25×10^{-6}. All analyses use 580 GPL96 expression records. P/FDR values retain correction over all endpoints in their original families and were not recalculated for the six displayed comparisons. Points and connecting lines show correlation magnitude, not CIs. Removing overlap attenuates but preserves negative correlations. Deconvolution estimates composition and does not measure cell fractions directly. This figure is not independent patient-level replication

Supplementary Fig. S4



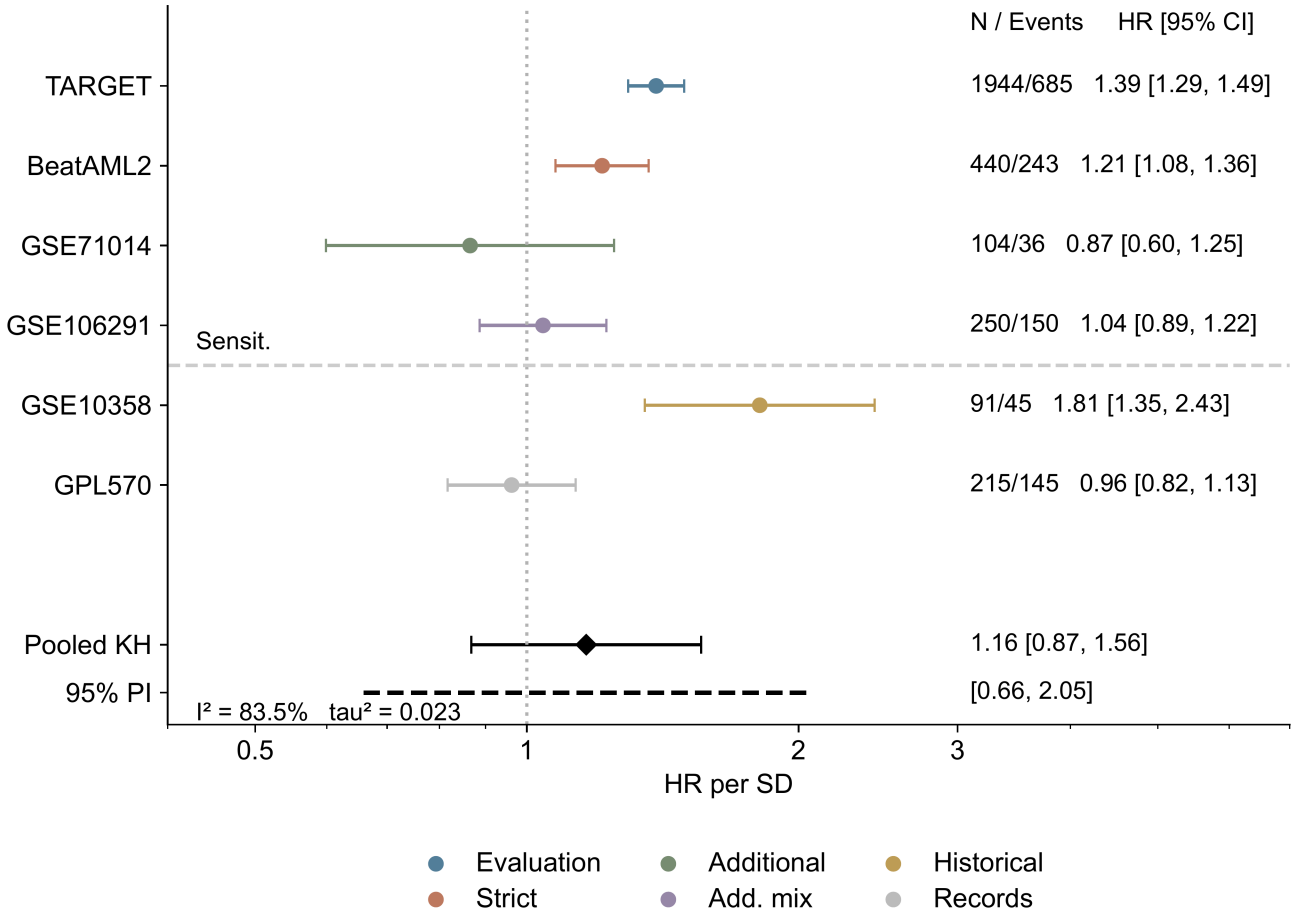
All pseudotime branches and endpoint state references. a LCDI–pseudotime Spearman rho for nine adult and four pediatric trajectories. Filled points indicate retained branches and open points non-retained adult branches; L identifiers are dataset-specific. Adult rho values follow the trajectory summaries' precision. Cell numbers are not independent patient sample sizes, and inferential significance is not shown. b Late-minus-early mean module scores for retained adult L1/L3/L4/L5 and pediatric L1–L4. Late and early endpoints are ≥90th and ≤10th pseudotime percentiles. Prog, Gran, Mono, and DC are project-curated progenitor, granulocytic, monocytic, and dendritic state references; complete genes are in Supplementary Table S10. These are not lineage-tracing measurements. No adult L3 samples and one pediatric L3 sample were shared between early and late endpoints; b is therefore descriptive, without paired patient inference. Adult branch selection used LCDI correlation strength. Displaying all branches discloses this selection context and does not constitute post-selection confirmatory validation. Ad/Pe, adult/pediatric

Supplementary Fig. S5



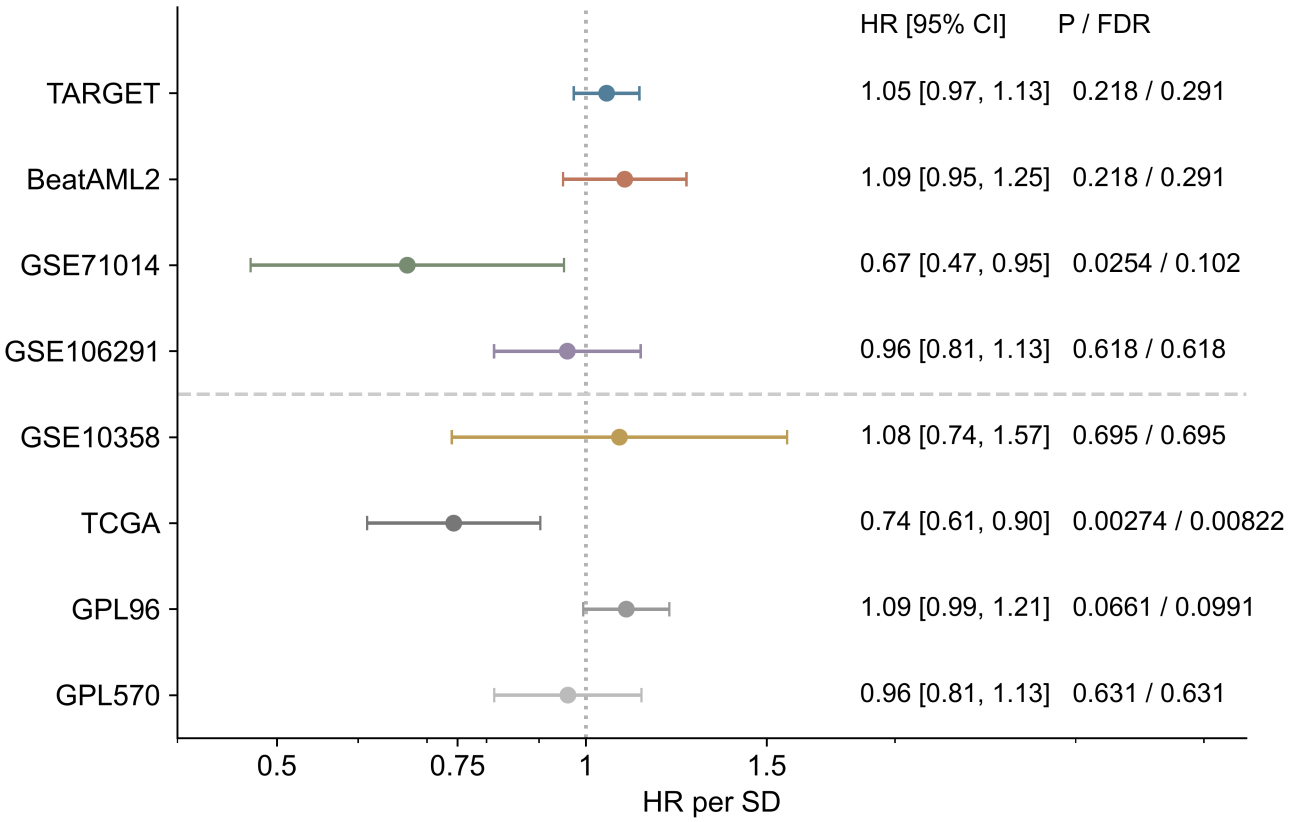
TF activity programs and sensitivity to adult control sources. a All 72 available cohort×method×TF comparisons for the predefined MYC/E2F and IFN/STAT/IRF programs, showing standardized source effects. Red/blue denotes higher/lower disease-source activity relative to comparison sources; gray indicates unavailable values. MYC/E2F effects are negative in 34/38 comparisons and IFN effects positive in 29/34. Adults comprise 16 D0 cases and four normal donors; pediatric data comprise 21 disease-source and 21 microenvironment-source units, not two independent patient groups. These are not LCDI high-versus-low comparisons. b Twenty-one direction reversals among 170 sensitivity results after requiring at least 20 cells in adult control sources or omitting BM1–BM4 donors individually. The counting unit is TF×method×scenario, not 21 independent TFs. No new inferential tests are shown. RNA-inferred activity is not measured protein activity. ULM, decoupleR univariate linear model; VIP, VIPER; SCN, SCENIC; Ad/Pe, adult/pediatric

Supplementary Fig. S6



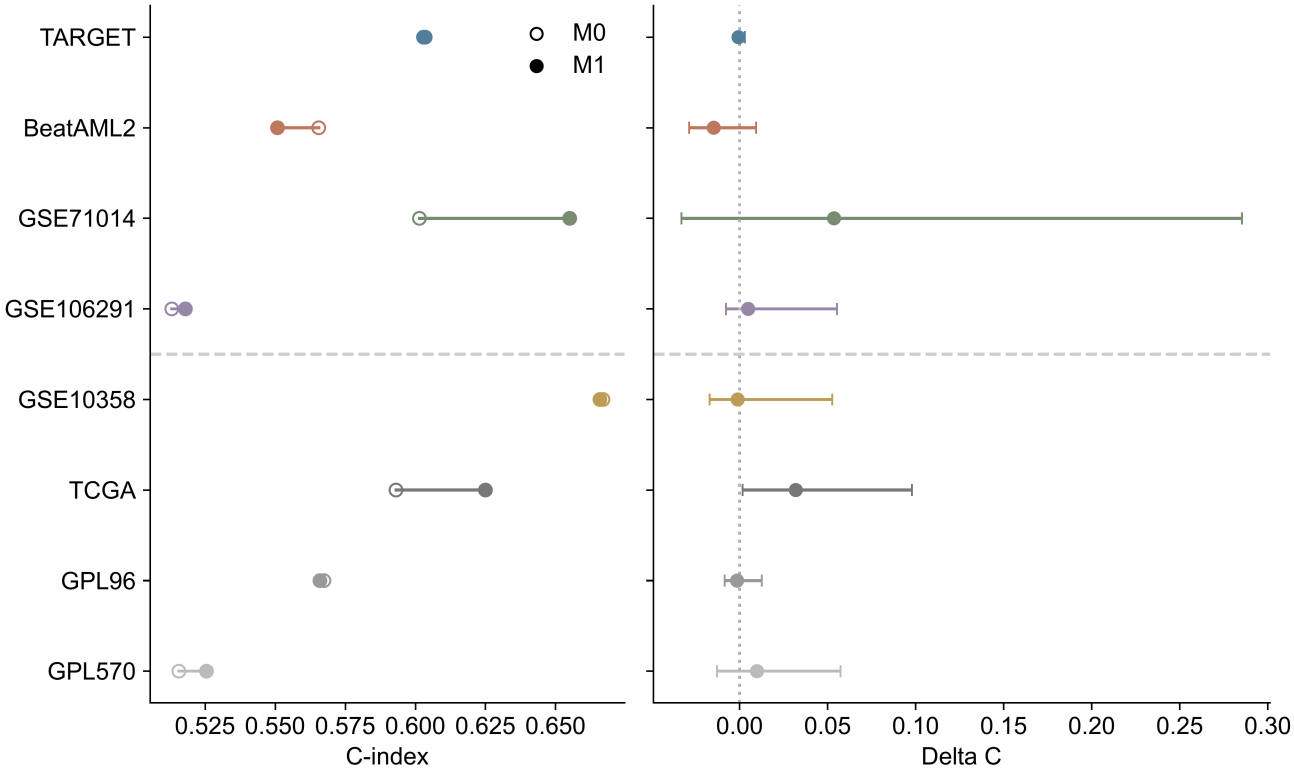
Overall survival effects per SD of frozen LCDI. The four primary evaluation cohorts are TARGET, BeatAML2, GSE71014, and GSE106291. Supplementary and expression-record cohorts below do not contribute to the displayed pooled value. Cohort lines show consistently defined Cox Wald 95% CIs; the diamond shows the REML pooled HR and Knapp–Hartung CI, and the dashed line the corresponding 95% prediction interval. Strict denotes primary independent BeatAML2 replication; Additional denotes further evaluation (including incompletely confirmed source independence in GSE106291). GPL570 is analyzed at the expression-record level. Development cohorts do not enter this meta-analysis. CI, confidence interval; HR, hazard ratio; SD, standard deviation; PI, prediction interval; KH, Knapp–Hartung; N/events, records or patients/deaths. Other pooling scenarios are in random_effects_meta_audited.tsv. The named auxiliary statistics files are supplied in Online Resource 2, folder 09_Additional_statistics

Supplementary Fig. S7



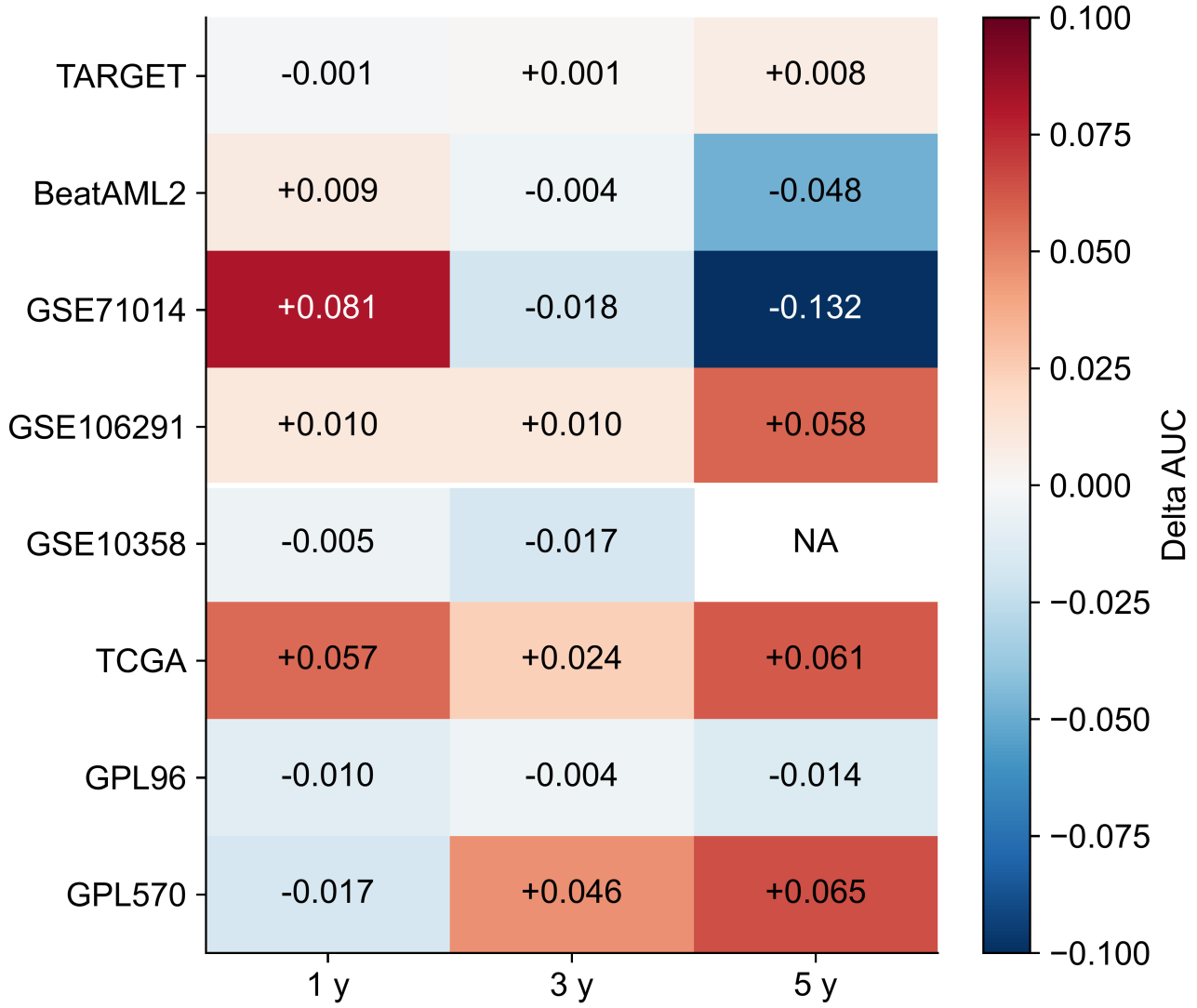
Conditional HR per SD of the frozen, combined non-MPO component after adjustment for MPO-risk. The four external Wald P values form one BH family; the four LRTs form another. Supplementary evaluation, development, and expression-record sensitivities below the dashed line should not be equated with external replication. Non-MPO denotes the frozen weighted sum of CREB1, SLC16A4, GPX4, and CASP1. MPO-risk is negatively oriented MPO scaled by its SD; the MPO component is −0.2095×Z(MPO) under the original scoring branch. FDR, false discovery rate; P, two-sided Wald probability

Supplementary Fig. S8



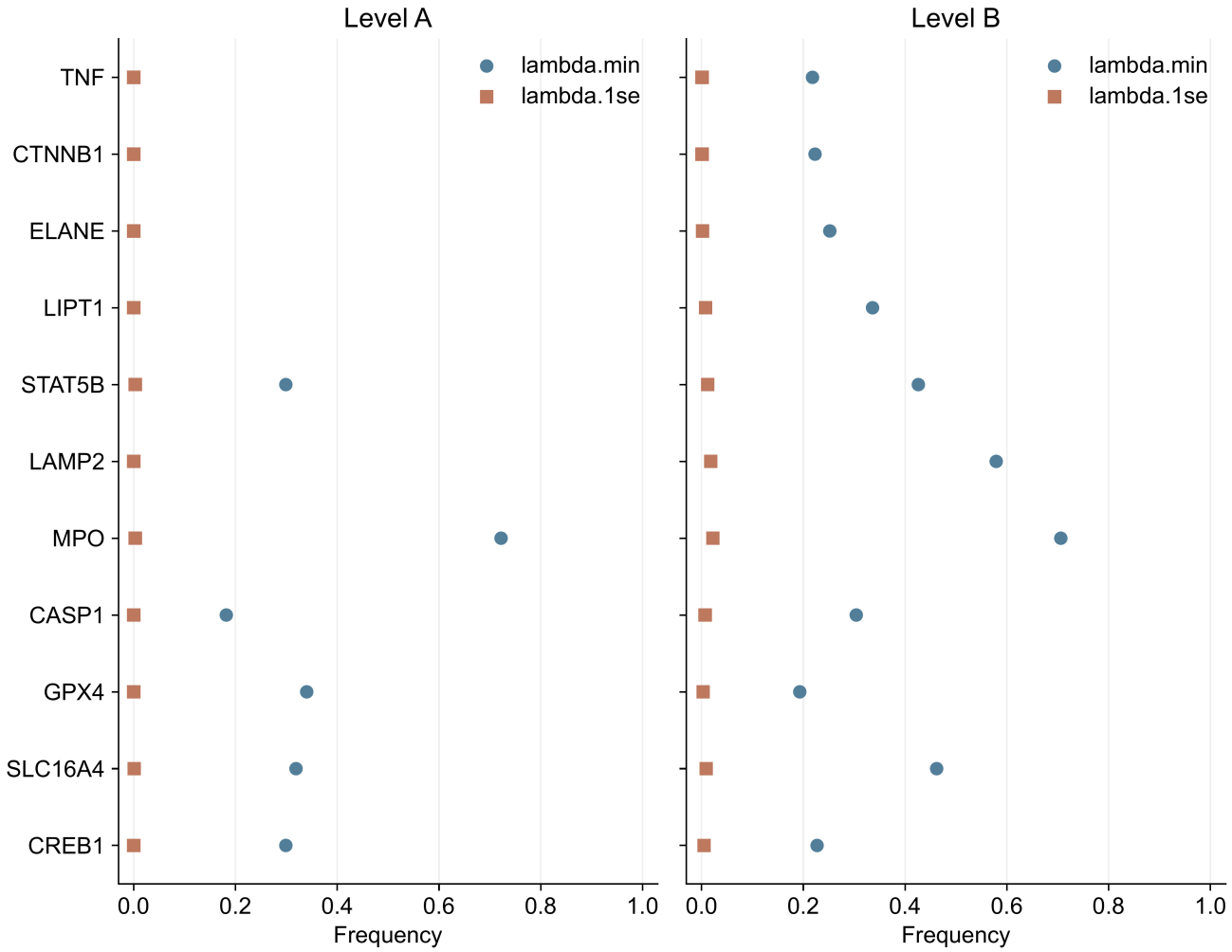
M0=Surv ~ MPO-risk; M1=M0 + the frozen combined non-MPO component. Open/filled points show apparent fitted C-indices for M0/M1 within each cohort. Paired delta C=M1−M0 and percentile intervals from 1,000 patient/record bootstrap samples appear on the right. Both models are refitted in identical resamples while gene standardization parameters remain fixed. Metrics use fitted linear predictors; M0 orientation can change with its fitted coefficient and cannot be substituted for the fixed MPO-risk C-index in Fig. 1. M0/M1 refit models of the scores; the five-gene formula remains fixed. Optimism-corrected delta C values are in paired_increment.tsv. The named auxiliary statistics files are supplied in Online Resource 2, folder 09_Additional_statistics

Supplementary Fig. S9



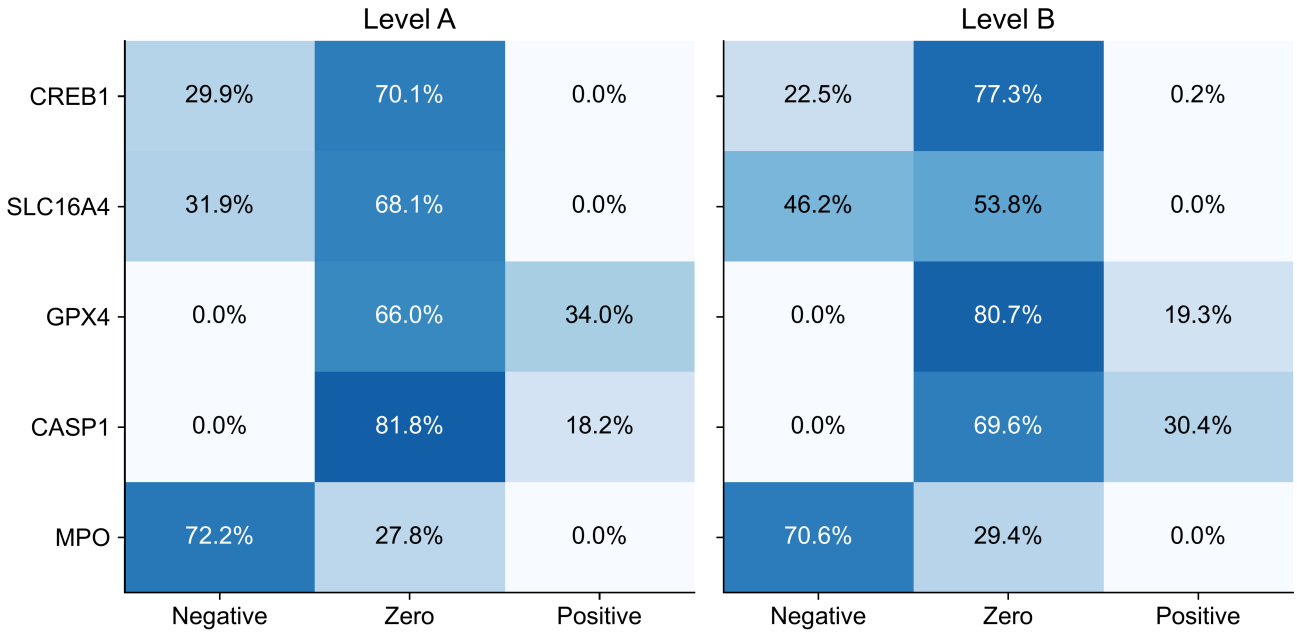
Paired delta AUC=M1−M0 at 1, 3, and 5 years, calculated by marginal IPCW timeROC using 365.25 days per year. All 1,000 resamples share patient/expression-record indices. Full intervals and valid iteration counts are in paired_increment.tsv. NA indicates non-estimable values; * indicates an available point estimate with fewer than 95% valid resamples, for which no stable interval is reported. Colors are clipped at ±0.10 for comparison, while cell labels retain actual values. AUC, area under the time-dependent ROC curve; IPCW, inverse probability of censoring weighting. The named auxiliary statistics files are supplied in Online Resource 2, folder 09_Additional_statistics

Supplementary Fig. S10



Level A fixes 17 candidates; level B repeats survival-dependent screening from the fixed set of 89 candidates. Each level uses 1,000 TCGA patient bootstrap samples, keeping repeated copies of each patient in the same fold during tenfold cross-validation. Circles/squares indicate lambda.min/lambda.1se. The frozen five genes and other genes with the highest selection frequencies in level B are shown. The analysis assesses stability of TCGA feature selection

Supplementary Fig. S11

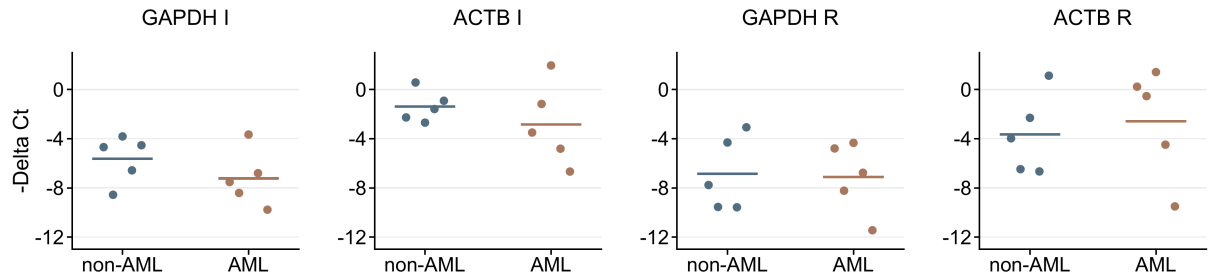


Negative/zero/positive coefficient frequencies at lambda.min in the two resampling levels, using all 1,000 iterations as the denominator. Zero includes non-selection. The figure shows TCGA LASSO coefficient directions, which must be distinguished from frozen GPL96 Cox coefficient directions. Complete lambda.1se and all-candidate frequencies, coefficient medians/interquartile ranges, and coselection results are in selection_frequency_all_genes.tsv, coefficient_stability.tsv, and alternative_gene_coselection.tsv, respectively. IQR, interquartile range; CV, cross-validation. The named auxiliary statistics files are supplied in Online Resource 2, folder 09_Additional_statistics

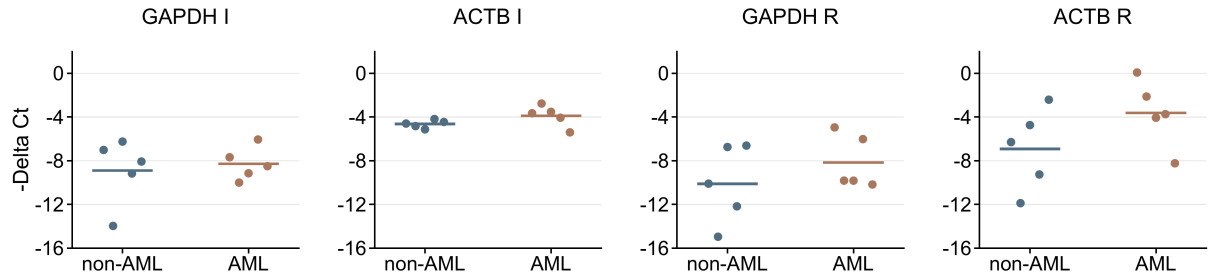
Sensit., sensitivity; Add. mix, additional evaluation with mixed source provenance.

Supplementary Fig. S12

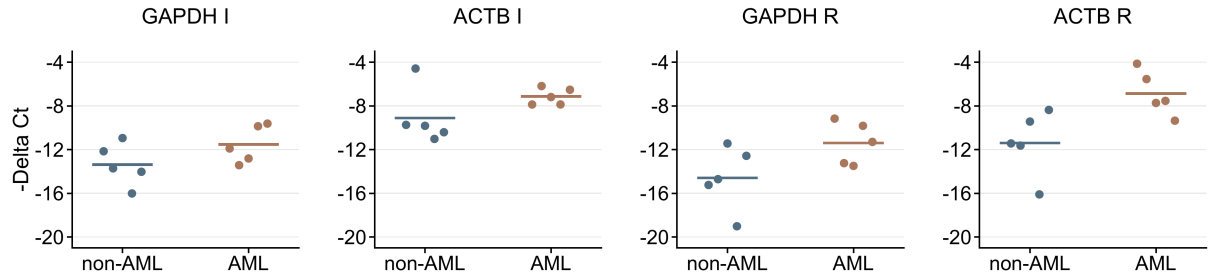
a **MPO**



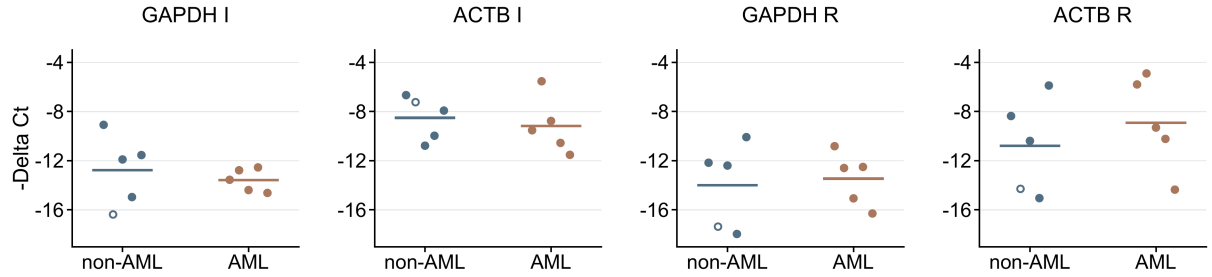
b **CREB1**



c **GPX4**



d **SLC16A4**



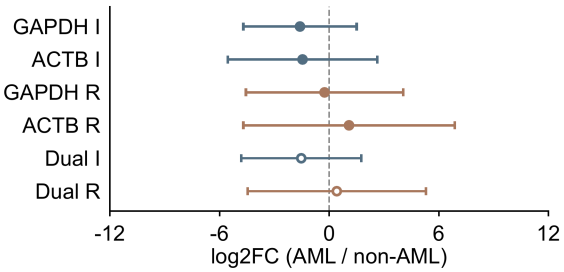
e **CASP1**

Ct available: non-AML 1/5; AML 1/5
Mean target Ct: 37.46 / 32.94
No group test or interval

Patient-level log expression distributions in clinical bone marrow RT-qPCR. a–d MPO, CREB1, GPX4, and SLC16A4. Each column represents a reference-normalization scheme; each point is one independent individual (five non-AML and five AML), with horizontal lines showing group means. Three technical replicate Ct values are averaged within individuals before calculating ΔCt=mean target Ct−mean reference Ct. The vertical axis is −ΔCt (log2 normalized expression), increasing with expression. Normalization schemes share the vertical scale within each gene; absolute values should not be directly compared across genes as measures of abundance. Open points mark mean target Ct≥35, indicating high Ct rather than an experimentally established quantification limit. e CASP1 has target Ct records for only one individual per group; other records are missing, are not imputed as zero or a fixed Ct, and are not tested between groups. Presence of a Ct record does not establish reliable quantification. I, initial reference assay; R, reference reassay; Ct, cycle threshold; AML, acute myeloid leukemia. Columns use the same individuals, with no target-gene remeasurement; the non-AML group is not a uniformly healthy control group

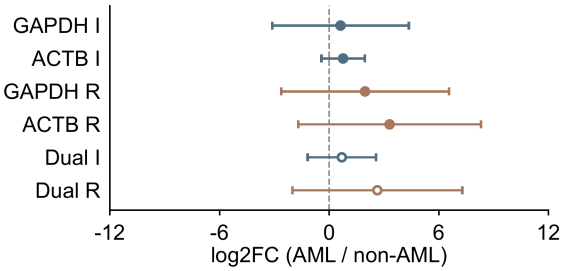
Supplementary Fig. S13

a MPO



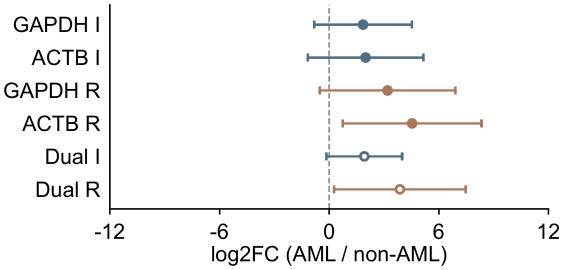
	log2FC [95% CI]	P	FDR
GAPDH I	-1.60 [-4.70,+1.50]	0.266	0.532
ACTB I	-1.46 [-5.55,+2.62]	0.403	0.537
GAPDH R	-0.26 [-4.56,+4.04]	0.893	0.893
ACTB R	+1.08 [-4.70,+6.86]	0.674	0.674
Dual I	-1.53 [-4.82,+1.75]	0.273	0.500
Dual R	+0.41 [-4.47,+5.29]	0.850	0.850

b CREB1



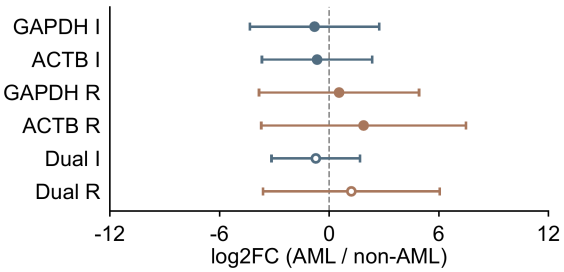
	log2FC [95% CI]	P	FDR
GAPDH I	+0.62 [-3.12,+4.35]	0.699	0.699
ACTB I	+0.76 [-0.43,+1.94]	0.162	0.328
GAPDH R	+1.96 [-2.63,+6.55]	0.347	0.694
ACTB R	+3.30 [-1.70,+8.31]	0.166	0.331
Dual I	+0.69 [-1.18,+2.55]	0.400	0.500
Dual R	+2.63 [-2.02,+7.28]	0.223	0.447

c GPX4



	log2FC [95% CI]	P	FDR
GAPDH I	+1.85 [-0.82,+4.52]	0.149	0.532
ACTB I	+1.99 [-1.17,+5.15]	0.164	0.328
GAPDH R	+3.19 [-0.52,+6.91]	0.082	0.326
ACTB R	+4.53 [+0.74,+8.33]	**0.026**	0.102
Dual I	+1.92 [-0.15,+3.99]	0.064	0.255
Dual R	+3.86 [+0.26,+7.46]	**0.039**	0.157

d SLC16A4



	log2FC [95% CI]	P	FDR
GAPDH I	-0.81 [-4.35,+2.74]	0.581	0.699
ACTB I	-0.67 [-3.68,+2.35]	0.622	0.622
GAPDH R	+0.54 [-3.84,+4.92]	0.779	0.893
ACTB R	+1.88 [-3.72,+7.48]	0.461	0.615
Dual I	-0.74 [-3.16,+1.69]	0.500	0.500
Dual R	+1.21 [-3.63,+6.04]	0.577	0.770

e CASP1

Only 1/5 per group has target Ct
No group effect, CI, P value or FDR is estimable

Between-group qPCR effects and reference-normalization sensitivity. a–d AML-versus-non-AML effects for MPO, CREB1, GPX4, and SLC16A4 under six schemes. Points show log2FC=−[mean(ΔCt_AML)−mean(ΔCt_non-AML)]; lines show Welch–Satterthwaite 95% CIs. Zero indicates no group difference and positive values higher AML expression. Each testable gene includes five independent individuals per group. Two-sided Welch P values on individual ΔCt values and BH FDRs appear on the right. Correction is performed across the four testable genes separately within each normalization scheme, without pooling across schemes. Bold $P<0.05$ indicates nominal significance only; no results reach $FDR<0.05$. Full precision, exact Mann–Whitney sensitivity results, and test families are in qPCR_full_statistics.tsv. e CASP1 has Ct records for only one individual per group; effect intervals and inferential statistics are not provided. I/R, initial/repeated reference assays; Dual, geometric-mean ACTB/GAPDH normalization as an exploratory sensitivity analysis; CI, confidence interval; FC, fold change; BH, Benjamini–Hochberg; FDR, false discovery rate. Blue/brown indicates initial/repeated references, and open points dual-reference schemes. All six schemes share individuals and target measurements and are not six independent replications. GPX4 is higher in AML under all six schemes. CREB1 is also consistently higher but opposite to its lower-expression direction in the same-platform GSE13159 reference; current data do not support directional replication. MPO and SLC16A4 directions depend on normalization. Panels a–d are arranged vertically in the order MPO, CREB1, GPX4, and SLC16A4. The cited qPCR_full_statistics.tsv is supplied in Online Resource 2, folder 09_Additional_statistics

Supplementary tables

All source values are preserved as text, including full numerical precision and empty cells. Column labels retain their source names. Row is the one-based source-data row and permits exact alignment across column blocks; repeated identifiers are for navigation only. S5a gives the candidate list, S5b the evidence groups, and S5c their references.

Table S1 Adult patient associations

Column block 1 of 4

Row	endpoint id	predictor	dataset	endpoint	family	n	rho
1	MEIS1	LCDI_z	adult	MEIS1	literature_representative_gene	16	0.1339220377279467
2	MEIS1	MPO_risk_z	adult	MEIS1	literature_representative_gene	16	0.3664460153215245
3	NRIP1	LCDI_z	adult	NRIP1	literature_representative_gene	16	0.611764705882353
4	NRIP1	MPO_risk_z	adult	NRIP1	literature_representative_gene	16	0.38235294117647056
5	MSI2	LCDI_z	adult	MSI2	literature_representative_gene	16	0.4147058823529412
6	MSI2	MPO_risk_z	adult	MSI2	literature_representative_gene	16	0.2529411764705882
7	ELANE	LCDI_z	adult	ELANE	literature_representative_gene	16	-0.36873316771508197
8	ELANE	MPO_risk_z	adult	ELANE	literature_representative_gene	16	-0.36283343703164056
9	AZU1	LCDI_z	adult	AZU1	literature_representative_gene	16	-0.32647058823529407
10	AZU1	MPO_risk_z	adult	AZU1	literature_representative_gene	16	-0.22647058823529412
11	LYZ	LCDI_z	adult	LYZ	literature_representative_gene	16	-0.22058823529411767
12	LYZ	MPO_risk_z	adult	LYZ	literature_representative_gene	16	-0.07941176470588236
13	MNDA	LCDI_z	adult	MNDA	literature_representative_gene	16	-0.2764705882352941
14	MNDA	MPO_risk_z	adult	MNDA	literature_representative_gene	16	-0.19117647058823528
15	CD14	LCDI_z	adult	CD14	literature_representative_gene	16	-0.002963044254729799
16	CD14	MPO_risk_z	adult	CD14	literature_representative_gene	16	0.005926088509459598

Column block 2 of 4

Row	endpoint id	predictor	p value	bootstrap ci low	bootstrap ci high	bootstrap valid	loo direction stability
1	MEIS1	LCDI_z	0.6209730849270084	-0.5274021212375227	0.6758315005794248	2000	0.875
2	MEIS1	MPO_risk_z	0.16271059713599237	-0.2697189727040473	0.7949979320817554	2000	1.0
3	NRIP1	LCDI_z	0.011788405053115212	0.08265131606390523	0.9335650383099339	2000	1.0

Row	endpoint id	predictor	p value	bootstrap ci low	bootstrap ci high	bootstrap valid	loo direction stability
4	NRIP1	MPO_risk_z	0.14385924954389048	-0.1981869268117245	0.7795372838190002	2000	1.0
5	MSI2	LCDI_z	0.11022940391770164	-0.13115853494804114	0.7444050318962339	2000	1.0
6	MSI2	MPO_risk_z	0.34456303133749866	-0.29568981481481477	0.6615080081116984	2000	1.0
7	ELANE	LCDI_z	0.15990335423430757	-0.8159275288588744	0.26494784112260233	2000	1.0
8	ELANE	MPO_risk_z	0.16721153262050986	-0.8187933729807865	0.2525009730381355	2000	1.0
9	AZU1	LCDI_z	0.2171594905792295	-0.6810830898617686	0.19478821837470345	2000	1.0
10	AZU1	MPO_risk_z	0.3989829266412891	-0.6313586790383198	0.29348258706467645	2000	1.0
11	LYZ	LCDI_z	0.4116664304979635	-0.6329983540941907	0.3124456486088853	2000	1.0
12	LYZ	MPO_risk_z	0.770025319039825	-0.4915370978361449	0.3962771506879423	2000	0.8125
13	MNDA	LCDI_z	0.2999421284923735	-0.6318310022065627	0.24910357321339321	2000	1.0
14	MNDA	MPO_risk_z	0.47816401281702825	-0.570349936402568	0.26328501754033656	2000	1.0
15	CD14	LCDI_z	0.9913106780700108	-0.5380346196124431	0.5888793765492586	2000	0.4375
16	CD14	MPO_risk_z	0.9826222715643256	-0.4870168831162919	0.536797235829943	2000	0.5

Column block 3 of 4

Row	endpoint id	predictor	loo min rho	loo max rho	self coupled	new panel	old family
1	MEIS1	LCDI_z	-0.05183201354778813	0.25558544611495526	False	True	False
2	MEIS1	MPO_risk_z	0.23571428571428568	0.5630029057776987	False	True	False
3	NRIP1	LCDI_z	0.5357142857142856	0.7285714285714285	False	True	False
4	NRIP1	MPO_risk_z	0.2714285714285714	0.5178571428571428	False	True	False
5	MSI2	LCDI_z	0.3178571428571428	0.5107142857142856	False	True	False
6	MSI2	MPO_risk_z	0.1357142857142857	0.36428571428571427	False	True	False
7	ELANE	LCDI_z	-0.49821108549919446	-0.23297640688811252	False	True	False
8	ELANE	MPO_risk_z	-0.48745832825820473	-0.225807902060786	False	True	False
9	AZU1	LCDI_z	-0.42857142857142855	-0.2071428571428571	False	True	False
10	AZU1	MPO_risk_z	-0.3857142857142856	-0.14285714285714282	False	True	False
11	LYZ	LCDI_z	-0.36428571428571427	-0.0964285714285714	False	True	False
12	LYZ	MPO_risk_z	-0.19999999999999998	0.017857142857142853	False	True	False

Row	endpoint id	predictor	loo min rho	loo max rho	self coupled	new panel	old family
13	MNDA	LCDI_z	-0.3928571428571428	-0.1857142857142857	False	True	False
14	MNDA	MPO_risk_z	-0.28571428571428564	-0.10357142857142855	False	True	False
15	CD14	LCDI_z	-0.1423481191058983	0.16845986344217367	False	True	True
16	CD14	MPO_risk_z	-0.11351812029964045	0.16129135861484714	False	True	True

Column block 4 of 4

Row	endpoint id	predictor	fdr union14	fdr	fdr old7
1	MEIS1	LCDI_z	0.7244685990815097	0.7096835256308668	
2	MEIS1	MPO_risk_z	0.39016024278118966	0.44589742032135965	
3	NRIP1	LCDI_z	0.08251883537180649	0.0943072404249217	
4	NRIP1	MPO_risk_z	0.39016024278118966	0.44589742032135965	
5	MSI2	LCDI_z	0.3086423309695646	0.4264089446248202	
6	MSI2	MPO_risk_z	0.535986937636109	0.637552017089371	
7	ELANE	LCDI_z	0.37310782654671765	0.4264089446248202	
8	ELANE	MPO_risk_z	0.39016024278118966	0.44589742032135965	
9	AZU1	LCDI_z	0.434318981158459	0.434318981158459	
10	AZU1	MPO_risk_z	0.5585760972978047	0.637552017089371	
11	LYZ	LCDI_z	0.6403700029968321	0.5488885739972846	
12	LYZ	MPO_risk_z	0.8382133641115262	0.8800289360455142	
13	MNDA	LCDI_z	0.5248987248616537	0.4799074055877976	
14	MNDA	MPO_risk_z	0.608572379948945	0.637552017089371	
15	CD14	LCDI_z	0.9913106780700108	0.9913106780700108	0.9913106780700108
16	CD14	MPO_risk_z	0.9826222715643256	0.9826222715643256	0.9826222715643256

Table S2 Pediatric patient associations

Column block 1 of 4

Row	endpoint id	predictor	dataset	endpoint	family	n	rho
1	MEIS1	LCDI_z	pediatric	MEIS1	literature_representative_gene	22	0.5619881613539803
2	MEIS1	MPO_risk_z	pediatric	MEIS1	literature_representative_gene	22	0.4495905290831843
3	NRIP1	LCDI_z	pediatric	NRIP1	literature_representative_gene	22	0.4206662902315077
4	NRIP1	MPO_risk_z	pediatric	NRIP1	literature_representative_gene	22	0.4647092038396387
5	MSI2	LCDI_z	pediatric	MSI2	literature_representative_gene	22	0.4387351778656127
6	MSI2	MPO_risk_z	pediatric	MSI2	literature_representative_gene	22	0.47826086956521746
7	ELANE	LCDI_z	pediatric	ELANE	literature_representative_gene	22	-0.6205533596837945
8	ELANE	MPO_risk_z	pediatric	ELANE	literature_representative_gene	22	-0.7651044607566347
9	AZU1	LCDI_z	pediatric	AZU1	literature_representative_gene	22	-0.3190287972896669
10	AZU1	MPO_risk_z	pediatric	AZU1	literature_representative_gene	22	-0.5155279503105591
11	LYZ	LCDI_z	pediatric	LYZ	literature_representative_gene	22	0.06832298136645965
12	LYZ	MPO_risk_z	pediatric	LYZ	literature_representative_gene	22	-0.009599096555618296
13	MNDA	LCDI_z	pediatric	MNDA	literature_representative_gene	22	-0.08413325804630155
14	MNDA	MPO_risk_z	pediatric	MNDA	literature_representative_gene	22	-0.1259175607001694
15	CD14	LCDI_z	pediatric	CD14	literature_representative_gene	22	0.0011296244449326236
16	CD14	MPO_risk_z	pediatric	CD14	literature_representative_gene	22	0.09601807781927302

Column block 2 of 4

Row	endpoint id	predictor	p value	bootstrap ci low	bootstrap ci high	bootstrap valid	loo direction stability
1	MEIS1	LCDI_z	0.006487553716785036	0.181006269838643	0.8340264203468128	2000	1.0
2	MEIS1	MPO_risk_z	0.035793984884606174	0.008773635593126635	0.8124879388164357	2000	1.0
3	NRIP1	LCDI_z	0.05123847389451633	-0.0180143760420475	0.708338371486405	2000	1.0
4	NRIP1	MPO_risk_z	0.029329953019594113	0.06756075707277633	0.720420372481946	2000	1.0
5	MSI2	LCDI_z	0.04108923308923487	5.517676865839918e-05	0.7339431721178805	2000	1.0
6	MSI2	MPO_risk_z	0.024357000698180267	0.08638190211222499	0.7441231055498102	2000	1.0
7	ELANE	LCDI_z	0.002060038149082111	-0.8810084854384684	-0.19190367842244985	2000	1.0

Row	endpoint id	predictor	p value	bootstrap ci low	bootstrap ci high	bootstrap valid	loo direction stability
8	ELANE	MPO_risk_z	3.354646476630029e-05	-0.9334906201435553	-0.4104506922951622	2000	1.0
9	AZU1	LCDI_z	0.14785060434145583	-0.6529181496777778	0.12167128948311433	2000	1.0
10	AZU1	MPO_risk_z	0.014063853357929122	-0.7822589918368474	-0.1013994230770693	2000	1.0
11	LYZ	LCDI_z	0.7625680708731801	-0.4288283614482025	0.5226104892070723	2000	0.8636363636363636
12	LYZ	MPO_risk_z	0.9661826982649286	-0.4596660975147031	0.44281002338440595	2000	0.5909090909090909
13	MNDA	LCDI_z	0.7097094861597419	-0.560270660102003	0.3840526216165058	2000	0.9090909090909091
14	MNDA	MPO_risk_z	0.5765977724368205	-0.5754433066473311	0.3333333333333333	2000	1.0
15	CD14	LCDI_z	0.9960192852974719	-0.4179044568716595	0.44884400253213724	2000	0.5454545454545454
16	CD14	MPO_risk_z	0.6707925860514529	-0.3687203147206047	0.5293136342202528	2000	1.0

Column block 3 of 4

Row	endpoint id	predictor	loo min rho	loo max rho	self coupled	new panel	old family
1	MEIS1	LCDI_z	0.5209483873325794	0.6532467532467533	False	True	False
2	MEIS1	MPO_risk_z	0.3890873865488967	0.5987012987012986	False	True	False
3	NRIP1	LCDI_z	0.3532467532467533	0.5441558441558442	False	True	False
4	NRIP1	MPO_risk_z	0.40129870129870127	0.5701298701298702	False	True	False
5	MSI2	LCDI_z	0.35454545454545455	0.509090909090909	False	True	False
6	MSI2	MPO_risk_z	0.4	0.574025974025974	False	True	False
7	ELANE	LCDI_z	-0.7233766233766233	-0.5662337662337662	False	True	False
8	ELANE	MPO_risk_z	-0.8337662337662338	-0.7298701298701298	False	True	False
9	AZU1	LCDI_z	-0.42207792207792205	-0.22597402597402597	False	True	False
10	AZU1	MPO_risk_z	-0.6337662337662338	-0.44675324675324674	False	True	False
11	LYZ	LCDI_z	-0.05454545454545454	0.18051948051948052	False	True	False
12	LYZ	MPO_risk_z	-0.14155844155844155	0.0961038961038961	False	True	False
13	MNDA	LCDI_z	-0.22857142857142856	0.0012987012987012987	False	True	False
14	MNDA	MPO_risk_z	-0.28051948051948056	-0.051948051948051945	False	True	False
15	CD14	LCDI_z	-0.08898993648948053	0.11428571428571428	False	True	True
16	CD14	MPO_risk_z	0.01104254686365817	0.23766233766233766	False	True	True

Column block 4 of 4

Row	endpoint id	predictor	fdr union14	fdr	fdr old7
1	MEIS1	LCDI_z	0.01990808900713047	0.025950214867140144	
2	MEIS1	MPO_risk_z	0.0626394735480608	0.05727037581536988	
3	NRIP1	LCDI_z	0.10247694778903267	0.10247694778903266	
4	NRIP1	MPO_risk_z	0.05865990603918823	0.05727037581536988	
5	MSI2	LCDI_z	0.0958748772082147	0.10247694778903266	
6	MSI2	MPO_risk_z	0.0568330016290873	0.05727037581536988	
7	ELANE	LCDI_z	0.009613511362383184	0.016480305192656888	
8	ELANE	MPO_risk_z	0.00023482525336410203	0.0002683717181304023	
9	AZU1	LCDI_z	0.2587385575975477	0.23656096694632933	
10	AZU1	MPO_risk_z	0.03937878940220154	0.056255413431716486	
11	LYZ	LCDI_z	0.8294565068186365	0.8715063667122058	
12	LYZ	MPO_risk_z	0.9661826982649286	0.9661826982649286	
13	MNDA	LCDI_z	0.8294565068186365	0.8715063667122058	
14	MNDA	MPO_risk_z	0.7825913503933616	0.7666200983445176	
15	CD14	LCDI_z	0.996019285297472	0.9960192852974719	0.996019285297472
16	CD14	MPO_risk_z	0.7825913503933616	0.7666200983445176	0.7825913503933616

Table S3 Pathway abbreviations

Row	abbreviation	full name
1	IFNa-resp	interferon alpha response
2	MYC-V1	MYC targets V1
3	IFNg-resp	interferon gamma response
4	ROS	reactive oxygen species pathway
5	UPR	unfolded protein response
6	NegMyelDif	negative regulation of myeloid cell differentiation

Row	abbreviation	full name
7	HemoReg	regulation of hemopoiesis
8	BactDef	defense response to bacterium
9	HumImm	humoral immune response
10	AntimHum	antimicrobial humoral response
11	SecGranLum	secretory granule lumen
12	CytVesLum	cytoplasmic vesicle lumen
13	VesLum	vesicle lumen
14	SpecGrLum	specific granule lumen
15	SpecGran	specific granule
16	ChemR-bind	chemokine receptor binding
17	CytoR-bind	cytokine receptor binding
18	SerEndoPep	serine-type endopeptidase activity
19	SerPep	serine-type peptidase activity
20	SerHydrol	serine hydrolase activity
21	ChemSig	KEGG_CHEMOKINE_SIGNALING_PATHWAY
22	CytoR-int	KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACTION
23	Asthma	KEGG_ASTHMA
24	RenAngSys	KEGG_RENIN_ANGIOTENSIN_SYSTEM
25	PPP	KEGG_PENTOSE_PHOSPHATE_PATHWAY
26	CytoTrans	cytoplasmic translation
27	Transl	translation
28	MacroCat	macromolecule catabolic process
29	PosGEreg	positive regulation of gene expression
30	NegGEreg	negative regulation of gene expression
31	NucCat	nucleobase-containing compound catabolic process
32	ProlifReg	regulation of cell population proliferation
33	BiotResp	response to external biotic stimulus

Row	abbreviation	full name
34	ProtOrg	protein-containing complex organization
35	Transport	intracellular transport
36	OrgResp	response to other organism
37	Ribosome	ribosome

Table S4 TF program direction counts

Row	cohort	method	program	available	higher	lower
1	Adult	SCENIC	IFN/STAT/IRF	4	3	1
2	Adult	SCENIC	MYC/E2F	4	2	2
3	Adult	VIPER	IFN/STAT/IRF	6	6	0
4	Adult	VIPER	MYC/E2F	7	0	7
5	Adult	decoupleR	IFN/STAT/IRF	6	4	2
6	Adult	decoupleR	MYC/E2F	7	0	7
7	Pediatric	SCENIC	IFN/STAT/IRF	6	5	1
8	Pediatric	SCENIC	MYC/E2F	6	2	4
9	Pediatric	VIPER	IFN/STAT/IRF	6	5	1
10	Pediatric	VIPER	MYC/E2F	7	0	7
11	Pediatric	decoupleR	IFN/STAT/IRF	6	6	0
12	Pediatric	decoupleR	MYC/E2F	7	0	7

Table S5a Candidate genes and literature annotations

Column block 1 of 3

Row	Gene symbol	Index	Combined list category	Original script group	Lactylation list member
1	EP300	1	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
2	HDAC1	2	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
3	HDAC2	3	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
4	HDAC3	4	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
5	SIRT1	5	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1

Row	Gene symbol	Index	Combined list category	Original script group	Lactylation list member
6	SIRT2	6	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
7	SIRT3	7	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
8	SIRT6	8	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
9	KAT2A	9	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
10	KAT2B	10	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
11	KAT5	11	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
12	KAT6A	12	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
13	KAT6B	13	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
14	KAT7	14	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
15	KAT8	15	Lactylation/lactate metabolism-related candidate	Lactylation-modifying enzyme (original script label)	1
16	LDHA	16	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
17	LDHB	17	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
18	LDHC	18	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
19	SLC16A1	19	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
20	SLC16A3	20	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
21	SLC16A4	21	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
22	PKM	22	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
23	PDHA1	23	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
24	PDHB	24	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
25	PDHX	25	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
26	GLO1	26	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
27	HAGH	27	Lactylation/lactate metabolism-related candidate	Lactate metabolism-related	1
28	HSPA2	28	Lactylation/lactate metabolism-related candidate	Heat shock protein	1
29	HSPA8	29	Lactylation/lactate metabolism-related candidate	Heat shock protein	1
30	HSP90AA1	30	Lactylation/lactate metabolism-related candidate	Heat shock protein	1
31	HSP90AB1	31	Lactylation/lactate metabolism-related candidate	Heat shock protein	1
32	HIF1A	32	Lactylation/lactate metabolism-related candidate	Transcription factor	1

Row	Gene symbol	Index	Combined list category	Original script group	Lactylation list member
33	MYC	33	Lactylation/lactate metabolism-related candidate	Transcription factor	1
34	MAX	34	Lactylation/lactate metabolism-related candidate	Transcription factor	1
35	STAT3	35	Lactylation/lactate metabolism-related candidate	Transcription factor	1
36	STAT5A	36	Lactylation/lactate metabolism-related candidate	Transcription factor	1
37	STAT5B	37	Lactylation/lactate metabolism-related candidate	Transcription factor	1
38	TP53	38	Lactylation/lactate metabolism-related candidate	Transcription factor	1
39	RELA	39	Lactylation/lactate metabolism-related candidate	Transcription factor	1
40	NFKB1	40	Lactylation/lactate metabolism-related candidate	Transcription factor	1
41	NFKB2	41	Lactylation/lactate metabolism-related candidate	Transcription factor	1
42	CREB1	42	Lactylation/lactate metabolism-related candidate	Transcription factor	1
43	JUN	43	Lactylation/lactate metabolism-related candidate	Transcription factor	1
44	FOS	44	Lactylation/lactate metabolism-related candidate	Transcription factor	1
45	RUNX1	45	Lactylation/lactate metabolism-related candidate	AML-related	1
46	CEBPA	46	Lactylation/lactate metabolism-related candidate	AML-related	1
47	FLT3	47	Lactylation/lactate metabolism-related candidate	AML-related	1
48	NPM1	48	Lactylation/lactate metabolism-related candidate	AML-related	1
49	DNMT3A	49	Lactylation/lactate metabolism-related candidate	AML-related	1
50	IDH1	50	Lactylation/lactate metabolism-related candidate	AML-related	1
51	IDH2	51	Lactylation/lactate metabolism-related candidate	AML-related	1
52	TET2	52	Lactylation/lactate metabolism-related candidate	AML-related	1
53	ASXL1	53	Lactylation/lactate metabolism-related candidate	AML-related	1
54	KMT2A	54	Lactylation/lactate metabolism-related candidate	AML-related	1
55	KIT	55	Lactylation/lactate metabolism-related candidate	AML-related	1
56	WT1	56	Lactylation/lactate metabolism-related candidate	AML-related	1
57	BCL2	57	PCD-related candidate	Apoptosis	0
58	BAX	58	PCD-related candidate	Apoptosis	0
59	BAK1	59	PCD-related candidate	Apoptosis	0

Row	Gene symbol	Index	Combined list category	Original script group	Lactylation list member
60	CASP3	60	PCD-related candidate	Apoptosis	0
61	CASP8	61	PCD-related candidate	Apoptosis	0
62	CASP9	62	PCD-related candidate	Apoptosis	0
63	FAS	63	PCD-related candidate	Apoptosis	0
64	TNF	64	PCD-related candidate	Apoptosis	0
65	ATG5	65	PCD-related candidate	Autophagy	0
66	ATG7	66	PCD-related candidate	Autophagy	0
67	ATG12	67	PCD-related candidate	Autophagy	0
68	BECN1	68	PCD-related candidate	Autophagy	0
69	SQSTM1	69	PCD-related candidate	Autophagy	0
70	MAP1LC3A	70	PCD-related candidate	Autophagy	0
71	MTOR	71	PCD-related candidate	Autophagy	0
72	GPX4	72	PCD-related candidate	Ferroptosis	0
73	SLC7A11	73	PCD-related candidate	Ferroptosis	0
74	ACSL4	74	PCD-related candidate	Ferroptosis	0
75	FTH1	75	PCD-related candidate	Ferroptosis	0
76	FTL	76	PCD-related candidate	Ferroptosis	0
77	TFRC	77	PCD-related candidate	Ferroptosis	0
78	GSDMD	78	PCD-related candidate	Pyroptosis	0
79	GSDME	79	PCD-related candidate	Pyroptosis	0
80	CASP1	80	PCD-related candidate	Pyroptosis	0
81	NLRP3	81	PCD-related candidate	Pyroptosis	0
82	AIM2	82	PCD-related candidate	Pyroptosis	0
83	MLKL	83	PCD-related candidate	Necroptosis	0
84	RIPK1	84	PCD-related candidate	Necroptosis	0
85	RIPK3	85	PCD-related candidate	Necroptosis	0
86	CYLD	86	PCD-related candidate	Necroptosis	0

Row	Gene symbol	Index	Combined list category	Original script group	Lactylation list member
87	FDX1	87	PCD-related candidate	Cuproptosis	0
88	LIAS	88	PCD-related candidate	Cuproptosis	0
89	LIPT1	89	PCD-related candidate	Cuproptosis	0
90	DLD	90	PCD-related candidate	Cuproptosis	0
91	DLAT	91	PCD-related candidate	Cuproptosis	0
92	CDH1	92	PCD-related candidate	Entosis	0
93	CTNNB1	93	PCD-related candidate	Entosis	0
94	RAC1	94	PCD-related candidate	Entosis	0
95	CDC42	95	PCD-related candidate	Entosis	0
96	MPO	96	PCD-related candidate	NETosis	0
97	ELANE	97	PCD-related candidate	NETosis	0
98	PADI4	98	PCD-related candidate	NETosis	0
99	NOX4	99	PCD-related candidate	NETosis	0
100	MAPK1	100	PCD-related candidate	Paraptosis	0
101	MAPK3	101	PCD-related candidate	Paraptosis	0
102	AKT1	102	PCD-related candidate	Paraptosis	0
103	PAK1	103	PCD-related candidate	Methuosis	0
104	KEAP1	104	PCD-related candidate	Oxeiptosis	0
105	PGAM5	105	PCD-related candidate	Oxeiptosis	0
106	AIFM1	106	PCD-related candidate	Oxeiptosis	0
107	CTSB	107	PCD-related candidate	Lysosomal_death	0
108	CTSD	108	PCD-related candidate	Lysosomal_death	0
109	LAMP1	109	PCD-related candidate	Lysosomal_death	0
110	LAMP2	110	PCD-related candidate	Lysosomal_death	0
111	BRCA1	111	PCD-related candidate	Mitotic_catastrophe	0
112	ATM	112	PCD-related candidate	Mitotic_catastrophe	0
113	ATR	113	PCD-related candidate	Mitotic_catastrophe	0

Column block 2 of 3

Row	Gene symbol	PCD list member	All PCD script categories	EntrezID	Final five gene member
1	EP300	0		2033	0
2	HDAC1	0		3065	0
3	HDAC2	0		3066	0
4	HDAC3	0		8841	0
5	SIRT1	0		23411	0
6	SIRT2	0		22933	0
7	SIRT3	0		23410	0
8	SIRT6	0		51548	0
9	KAT2A	0		2648	0
10	KAT2B	0		8850	0
11	KAT5	0		10524	0
12	KAT6A	0		7994	0
13	KAT6B	0		23522	0
14	KAT7	0		11143	0
15	KAT8	0		84148	0
16	LDHA	0		3939	0
17	LDHB	0		3945	0
18	LDHC	0		3948	0
19	SLC16A1	0		6566	0
20	SLC16A3	0		9123	0
21	SLC16A4	0		9122	1
22	PKM	0		5315	0
23	PDHA1	0		5160	0

Row	Gene symbol	PCD list member	All PCD script categories	EntrezID	Final five gene member
24	PDHB	0		5162	0
25	PDHX	0		8050	0
26	GLO1	0		2739	0
27	HAGH	0		3029	0
28	HSPA2	0		3306	0
29	HSPA8	0		3312	0
30	HSP90AA1	0		3320	0
31	HSP90AB1	0		3326	0
32	HIF1A	0		3091	0
33	MYC	0		4609	0
34	MAX	0		4149	0
35	STAT3	0		6774	0
36	STAT5A	0		6776	0
37	STAT5B	0		6777	0
38	TP53	1	Apoptosis; Mitotic_catastrophe	7157	0
39	RELA	0		5970	0
40	NFKB1	0		4790	0
41	NFKB2	0		4791	0
42	CREB1	0		1385	1
43	JUN	0		3725	0
44	FOS	0		2353	0
45	RUNX1	0		861	0
46	CEBPA	0		1050	0
47	FLT3	0		2322	0

Row	Gene symbol	PCD list member	All PCD script categories	EntrezID	Final five gene member
48	NPM1	0		4869	0
49	DNMT3A	0		1788	0
50	IDH1	0		3417	0
51	IDH2	0		3418	0
52	TET2	0		54790	0
53	ASXL1	0		171023	0
54	KMT2A	0		4297	0
55	KIT	0		3815	0
56	WT1	0		7490	0
57	BCL2	1	Apoptosis	596	0
58	BAX	1	Apoptosis	581	0
59	BAK1	1	Apoptosis	578	0
60	CASP3	1	Apoptosis	836	0
61	CASP8	1	Apoptosis	841	0
62	CASP9	1	Apoptosis	842	0
63	FAS	1	Apoptosis	355	0
64	TNF	1	Apoptosis	7124	0
65	ATG5	1	Autophagy	9474	0
66	ATG7	1	Autophagy	10533	0
67	ATG12	1	Autophagy	9140	0
68	BECN1	1	Autophagy	8678	0
69	SQSTM1	1	Autophagy	8878	0
70	MAP1LC3A	1	Autophagy	84557	0
71	MTOR	1	Autophagy; Paraptosis	2475	0
72	GPX4	1	Ferroptosis	2879	1
73	SLC7A11	1	Ferroptosis	23657	0

Row	Gene symbol	PCD list member	All PCD script categories	EntrezID	Final five gene member
74	ACSL4	1	Ferroptosis	2182	0
75	FTH1	1	Ferroptosis	2495	0
76	FTL	1	Ferroptosis	2512	0
77	TFRC	1	Ferroptosis	7037	0
78	GSDMD	1	Pyroptosis	79792	0
79	GSDME	1	Pyroptosis	1687	0
80	CASP1	1	Pyroptosis	834	1
81	NLRP3	1	Pyroptosis	114548	0
82	AIM2	1	Pyroptosis	9447	0
83	MLKL	1	Necroptosis	197259	0
84	RIPK1	1	Necroptosis	8737	0
85	RIPK3	1	Necroptosis	11035	0
86	CYLD	1	Necroptosis	1540	0
87	FDX1	1	Cuproptosis	2230	0
88	LIAS	1	Cuproptosis	11019	0
89	LIPT1	1	Cuproptosis	51601	0
90	DLD	1	Cuproptosis	1738	0
91	DLAT	1	Cuproptosis	1737	0
92	CDH1	1	Entosis	999	0
93	CTNNB1	1	Entosis	1499	0
94	RAC1	1	Entosis; Methuosis	5879	0
95	CDC42	1	Entosis; Methuosis	998	0
96	MPO	1	NETosis	4353	1
97	ELANE	1	NETosis	1991	0
98	PADI4	1	NETosis	23569	0
99	NOX4	1	NETosis	50507	0
100	MAPK1	1	Paraptosis	5594	0

Row	Gene symbol	PCD list member	All PCD script categories	EntrezID	Final five gene member
101	MAPK3	1	Paraptosis	5595	0
102	AKT1	1	Paraptosis	207	0
103	PAK1	1	Methuosis	5058	0
104	KEAP1	1	Oxeiptosis	9817	0
105	PGAM5	1	Oxeiptosis	192111	0
106	AIFM1	1	Oxeiptosis	9131	0
107	CTSB	1	Lysosomal_death	1508	0
108	CTSD	1	Lysosomal_death	1509	0
109	LAMP1	1	Lysosomal_death	3916	0
110	LAMP2	1	Lysosomal_death	3920	0
111	BRCA1	1	Mitotic_catastrophe	672	0
112	ATM	1	Mitotic_catastrophe	472	0
113	ATR	1	Mitotic_catastrophe	545	0

Column block 3 of 3

Row	Gene symbol	Literature support level	Evidence group	Reference ID	Annotation date
1	EP300	A	E01	R01	2026-10-07
2	HDAC1	A	E02	R02	2026-10-07
3	HDAC2	A	E02	R02	2026-10-07
4	HDAC3	A	E02	R02	2026-10-07
5	SIRT1	A	E03	R02	2026-10-07
6	SIRT2	A	E03	R02	2026-10-07
7	SIRT3	A	E03	R02	2026-10-07
8	SIRT6	C	U		2026-10-07
9	KAT2A	A	E27	R19	2026-10-07
10	KAT2B	C	U		2026-10-07
11	KAT5	A	E25	R17	2026-10-07

Row	Gene symbol	Literature support level	Evidence group	Reference ID	Annotation date
12	KAT6A	C	U		2026-10-07
13	KAT6B	C	U		2026-10-07
14	KAT7	A	E04	R03	2026-10-07
15	KAT8	A	E26	R18	2026-10-07
16	LDHA	B	E28	R20	2026-10-07
17	LDHB	B	E29	R21	2026-10-07
18	LDHC	B	E29	R21	2026-10-07
19	SLC16A1	B	E23	R16	2026-10-07
20	SLC16A3	B	E23	R16	2026-10-07
21	SLC16A4	B	E24	R16	2026-10-07
22	PKM	C	U		2026-10-07
23	PDHA1	B	E30	R08	2026-10-07
24	PDHB	B	E30	R08	2026-10-07
25	PDHX	A	E05	R04	2026-10-07
26	GLO1	B	E06	R05	2026-10-07
27	HAGH	B	E06	R05	2026-10-07
28	HSPA2	C	U		2026-10-07
29	HSPA8	C	U		2026-10-07
30	HSP90AA1	C	U		2026-10-07
31	HSP90AB1	C	U		2026-10-07
32	HIF1A	C	U		2026-10-07
33	MYC	B	E28	R20	2026-10-07
34	MAX	C	U		2026-10-07
35	STAT3	C	U		2026-10-07
36	STAT5A	A	E05	R04	2026-10-07
37	STAT5B	C	U		2026-10-07

Row	Gene symbol	Literature support level	Evidence group	Reference ID	Annotation date
38	TP53	B	E08	R07	2026-10-07
39	RELA	C	U		2026-10-07
40	NFKB1	C	U		2026-10-07
41	NFKB2	C	U		2026-10-07
42	CREB1	B	E22	R13	2026-10-07
43	JUN	C	U		2026-10-07
44	FOS	C	U		2026-10-07
45	RUNX1	B	E07	R06	2026-10-07
46	CEBPA	B	E07	R06	2026-10-07
47	FLT3	B	E07	R06	2026-10-07
48	NPM1	B	E07	R06	2026-10-07
49	DNMT3A	B	E07	R06	2026-10-07
50	IDH1	B	E07	R06	2026-10-07
51	IDH2	B	E07	R06	2026-10-07
52	TET2	B	E07	R06	2026-10-07
53	ASXL1	B	E07	R06	2026-10-07
54	KMT2A	B	E07	R06	2026-10-07
55	KIT	B	E07	R06	2026-10-07
56	WT1	B	E07	R06	2026-10-07
57	BCL2	B	E08	R07	2026-10-07
58	BAX	B	E08	R07	2026-10-07
59	BAK1	B	E08	R07	2026-10-07
60	CASP3	B	E08	R07	2026-10-07
61	CASP8	B	E08	R07	2026-10-07
62	CASP9	B	E08	R07	2026-10-07
63	FAS	B	E08	R07	2026-10-07
64	TNF	B	E08	R07	2026-10-07

Row	Gene symbol	Literature support level	Evidence group	Reference ID	Annotation date
65	ATG5	B	E09	R12	2026-10-07
66	ATG7	B	E09	R12	2026-10-07
67	ATG12	B	E09	R12	2026-10-07
68	BECN1	B	E09	R12	2026-10-07
69	SQSTM1	B	E09	R12	2026-10-07
70	MAP1LC3A	B	E09	R12	2026-10-07
71	MTOR	B	E09	R12	2026-10-07
72	GPX4	A	E10	R14	2026-10-07
73	SLC7A11	C	U		2026-10-07
74	ACSL4	B	E11	R07	2026-10-07
75	FTH1	C	U		2026-10-07
76	FTL	C	U		2026-10-07
77	TFRC	C	U		2026-10-07
78	GSDMD	B	E12	R07	2026-10-07
79	GSDME	B	E12	R07	2026-10-07
80	CASP1	A	E13	R15	2026-10-07
81	NLRP3	B	E12	R07	2026-10-07
82	AIM2	B	E12	R07	2026-10-07
83	MLKL	B	E14	R07	2026-10-07
84	RIPK1	B	E14	R07	2026-10-07
85	RIPK3	B	E14	R07	2026-10-07
86	CYLD	B	E14	R07	2026-10-07
87	FDX1	A	E15	R08	2026-10-07
88	LIAS	A	E15	R08	2026-10-07
89	LIPT1	A	E15	R08	2026-10-07
90	DLD	A	E15	R08	2026-10-07
91	DLAT	A	E15	R08	2026-10-07

Row	Gene symbol	Literature support level	Evidence group	Reference ID	Annotation date
92	CDH1	B	E16	R07	2026-10-07
93	CTNNB1	C	U		2026-10-07
94	RAC1	A	E17	R11	2026-10-07
95	CDC42	C	U		2026-10-07
96	MPO	A	E18	R10	2026-10-07
97	ELANE	A	E18	R10	2026-10-07
98	PADI4	B	E19	R07	2026-10-07
99	NOX4	C	U		2026-10-07
100	MAPK1	C	U		2026-10-07
101	MAPK3	C	U		2026-10-07
102	AKT1	C	U		2026-10-07
103	PAK1	C	U		2026-10-07
104	KEAP1	A	E20	R09	2026-10-07
105	PGAM5	A	E20	R09	2026-10-07
106	AIFM1	A	E20	R09	2026-10-07
107	CTSB	B	E21	R07	2026-10-07
108	CTSD	B	E21	R07	2026-10-07
109	LAMP1	C	U		2026-10-07
110	LAMP2	C	U		2026-10-07
111	BRCA1	C	U		2026-10-07
112	ATM	B	E31	R17	2026-10-07
113	ATR	C	U		2026-10-07

Table S5b Evidence groups and interpretation

Column block 1 of 2

Row	evidence group	genes	level	reference ids
1	E01	EP300	A	R01
2	E02	HDAC1 HDAC2 HDAC3	A	R02
3	E03	SIRT1 SIRT2 SIRT3	A	R02
4	E04	KAT7	A	R03
5	E05	STAT5A PDHX	A	R04
6	E06	GLO1 HAGH	B	R05
7	E07	RUNX1 CEBPA FLT3 NPM1 DNMT3A IDH1 IDH2 TET2 ASXL1 KMT2A KIT WT1	B	R06
8	E08	TP53 BCL2 BAX BAK1 CASP3 CASP8 CASP9 FAS TNF	B	R07
9	E09	ATG5 ATG7 ATG12 BECN1 SQSTM1 MAP1LC3A MTOR	B	R12
10	E10	GPX4	A	R14
11	E11	ACSL4	B	R07
12	E12	GSDMD GSDME NLRP3 AIM2	B	R07
13	E13	CASP1	A	R15
14	E14	MLKL RIPK1 RIPK3 CYLD	B	R07
15	E15	FDX1 LIAS LIPT1 DLD DLAT	A	R08
16	E16	CDH1	B	R07
17	E17	RAC1	A	R11
18	E18	MPO ELANE	A	R10
19	E19	PADI4	B	R07
20	E20	KEAP1 PGAM5 AIFM1	A	R09
21	E21	CTSB CTSD	B	R07
22	E22	CREB1	B	R13
23	E23	SLC16A1 SLC16A3	B	R16
24	E24	SLC16A4	B	R16
25	E25	KAT5	A	R17

Row	evidence group	genes	level	reference ids
26	E26	KAT8	A	R18
27	E27	KAT2A	A	R19
28	E28	LDHA MYC	B	R20
29	E29	LDHB LDHC	B	R21
30	E30	PDHA1 PDHB	B	R08
31	E31	ATM	B	R17

Column block 2 of 2

Row	evidence group	supported scope	evidence location	interpretation boundary
1	E01	Histone lactylation catalysis	Biochemical reconstitution and p300-dependent transcription	Does not establish all KAT family members as lactyltransferases.
2	E02	Histone lysine delactylation	Abstract; enzymatic assays; HDAC1/3 cellular experiments	Cellular confirmation in this study is stronger for HDAC1/3 than HDAC2.
3	E03	In-vitro lysine delactylation	Abstract and enzyme screen	In-vitro activity is not proof of AML-specific histone delactylation.
4	E04	HBO1-mediated histone lactylation	Figures 1-3; HBO1 is KAT7	Evidence from biochemical and cellular systems, not this AML cohort.
5	E05	AML STAT5A/E3BP-lactylation axis	STAT5A1*6 construct; E3BP nuclear localization experiments; E3BP is PDHX	STAT5A-specific experiments do not automatically establish STAT5B equivalence.
6	E06	Glyoxalase/lactoylglutathione context	Glyoxalase intermediate and non-enzymatic lactoylation; HAGH is glyoxalase II	Chemical lactoylation context; not evidence that these are L-lactyltransferases.
7	E07	AML genomic context	Recurrent alterations and molecular categories; KMT2A is MLL	AML relevance, not direct lactylation involvement.
8	E08	Apoptosis context	Intrinsic/extrinsic apoptosis sections; BAK is BAK1	No AML-specific execution claim.
9	E09	Autophagy machinery	Introduction; Figures 2-4; ATG8 sub-network	Autophagy association alone does not establish autophagy-dependent death.
10	E10	AML ferroptosis regulation	GPX4 inhibition experiments	Not proof that the LCDI coefficient measures ferroptosis activity.
11	E11	Ferroptosis context	Ferroptosis section	Mechanism annotation, not prognostic validation.
12	E12	Pyroptosis context	Pyroptosis section; AIM2-linked study cited therein	Sensor and effector roles differ.
13	E13	AML pyroptosis	CARD8/pro-caspase-1 response to DPP8/9 inhibition	State-associated expression does not by itself demonstrate pyroptosis.
14	E14	Necroptosis context	Necroptosis section	Not an AML-specific death assay.
15	E15	Copper-dependent death	CRISPR screening and lipoylation pathway; Figures 3-5	Copper-induced death in experimental systems; not evidence of activity in these AML patients.
16	E16	Entosis context	Entotic cell death section	Cell adhesion involvement; not evidence for every listed small GTPase.
17	E17	Methuosis context	Ras/Rac1/Arf6 functional experiments	Supports the secondary methuosis assignment; primary entosis assignment is not established by this paper.

Row	evidence group	supported scope	evidence location	interpretation boundary
18	E18	NET formation	Granule release and chromatin experiments	NET formation evidence does not establish NETotic death in AML blasts.
19	E19	NET formation context	NETotic cell death section	Stimulus-dependent involvement.
20	E20	Oxeiptosis	ROS-induced KEAP1-PGAM5-AIFM1 experiments	No inference that all roles of these proteins are specific to oxeiptosis.
21	E21	Lysosomal death context	Lysosome-dependent death section	Context annotation; not proof of pathway activation.
22	E22	Hematopoiesis and leukemogenesis	CREB perturbation in hematopoietic systems	Supports myeloid regulatory context, not direct lactylation catalysis.
23	E23	Lactate transport	MCT1 and MCT4 substrate annotations	SLC16A3 is canonical MCT4.
24	E24	Transporter-family context	MCT5 entry; orphan transporter comments	Not established as a lactate transporter; do not call canonical MCT4.
25	E25	TIP60-mediated NBS1 lactylation	Figure 2; TIP60 is KAT5	Experimental evidence in solid-tumour systems, not the AML cohorts.
26	E26	eEF1A2 lactylation	Figure 4; in-vitro and cellular experiments	Not AML-specific evidence.
27	E27	GCN5-linked histone lactylation	GCN5 recruitment and H3K18 lactylation experiments	Context is post-infarction monocytes, not AML.
28	E28	Transcriptional control of lactate metabolism	LDH-A promoter and c-Myc experiments	Supports metabolic-regulatory context, not direct lactylation catalysis.
29	E29	Lactate dehydrogenase isoenzyme context	Introduction and LDH isoenzyme scheme	Isoenzyme context; structure experiment concerns LDHA, not direct LDHB/LDHC lactylation.
30	E30	Pyruvate dehydrogenase/metabolic context	Seven CRISPR resistance hits; author manuscript page 4	Supports metabolic and copper-death relevance, not their original lactylation label.
31	E31	DNA-repair/lactylation signalling context	Extended Data Figure 7h,i; ATM depletion	Supports DNA-repair context, not the original mitotic-catastrophe assignment specifically.

Table S5c Candidate evidence references

Column block 1 of 2

Row	reference id	title	doi	url
1	R01	Zhang et al. Metabolic regulation of gene expression by histone lactylation. Nature. 2019;574:575-580.	10.1038/s41586-019-1678-1	https://pubmed.ncbi.nlm.nih.gov/31645732/
2	R02	Moreno-Yruela et al. Class I histone deacetylases (HDAC1-3) are histone lysine delactylases. Sci Adv. 2022;8:eabi6696.	10.1126/sciadv.abi6696	https://pubmed.ncbi.nlm.nih.gov/35044827/
3	R03	Niu et al. HBO1 catalyzes lysine lactylation and mediates histone H3K9la to regulate gene transcription. Nat Commun. 2024;15:3561.	10.1038/s41467-024-47900-6	https://www.nature.com/articles/s41467-024-47900-6
4	R04	Huang et al. STAT5 promotes PD-L1 expression by facilitating histone lactylation to drive immunosuppression in acute myeloid leukemia. Signal Transduct Target Ther. 2023;8:391.	10.1038/s41392-023-01605-2	https://www.nature.com/articles/s41392-023-01605-2

Row	reference id	title	doi	url
5	R05	Gaffney et al. Non-enzymatic Lysine Lactoylation of Glycolytic Enzymes. Cell Chem Biol. 2020;27:206-213.e6.	10.1016/j.chembiol.2019.11.005	https://pubmed.ncbi.nlm.nih.gov/31767537/
6	R06	Cancer Genome Atlas Research Network. Genomic and epigenomic landscapes of adult de novo acute myeloid leukemia. N Engl J Med. 2013;368:2059-2074.	10.1056/NEJMoa1301689	https://pubmed.ncbi.nlm.nih.gov/23634996/
7	R07	Galluzzi et al. Molecular mechanisms of cell death: recommendations of the Nomenclature Committee on Cell Death 2018. Cell Death Differ. 2018;25:486-541.	10.1038/s41418-017-0012-4	https://link.springer.com/article/10.1038/s41418-017-0012-4
8	R08	Tsvetkov et al. Copper induces cell death by targeting lipoylated TCA cycle proteins. Science. 2022;375:1254-1261.	10.1126/science.abf0529	https://pubmed.ncbi.nlm.nih.gov/35298263/
9	R09	Holze et al. Oxeiptosis, a ROS-induced caspase-independent apoptosis-like cell-death pathway. Nat Immunol. 2018;19:130-140.	10.1038/s41590-017-0013-y	https://www.nature.com/articles/s41590-017-0013-y
10	R10	Papayannopoulos et al. Neutrophil elastase and myeloperoxidase regulate the formation of neutrophil extracellular traps. J Cell Biol. 2010;191:677-691.	10.1083/jcb.201006052	https://pubmed.ncbi.nlm.nih.gov/20974816/
11	R11	Bhanot et al. Induction of nonapoptotic cell death by activated Ras requires inverse regulation of Rac1 and Arf6. Mol Cancer Res. 2010.		https://pubmed.ncbi.nlm.nih.gov/20713492/
12	R12	Behrends et al. Network organization of the human autophagy system. Nature. 2010;466:68-76.	10.1038/nature09204	https://pmc.ncbi.nlm.nih.gov/articles/PMC2901998/
13	R13	Cheng et al. CREB is a critical regulator of normal hematopoiesis and leukemogenesis. Blood. 2008;111:1182-1192.	10.1182/blood-2007-04-083600	https://doi.org/10.1182/blood-2007-04-083600
14	R14	Akiyama et al. Mitochondrial regulation of GPX4 inhibition-mediated ferroptosis in acute myeloid leukemia. Leukemia. 2024;38:729-740.	10.1038/s41375-023-02117-2	https://doi.org/10.1038/s41375-023-02117-2
15	R15	Johnson et al. DPP8/DPP9 inhibitor-induced pyroptosis for treatment of acute myeloid leukemia. Nat Med. 2018;24:1151-1156.	10.1038/s41591-018-0082-y	https://doi.org/10.1038/s41591-018-0082-y
16	R16	IUPHAR/BPS Guide to PHARMACOLOGY. SLC16 family of monocarboxylate transporters.		https://www.guidetopharmacology.org/GRAC/FamilyDisplayForward?familyId=188
17	R17	NBS1 lactylation is required for efficient DNA repair and chemotherapy resistance. Nature. 2024.	10.1038/s41586-024-07620-9	https://www.nature.com/articles/s41586-024-07620-9
18	R18	KAT8-catalyzed lactylation promotes eEF1A2-mediated protein synthesis and colorectal carcinogenesis. 2024.		https://pmc.ncbi.nlm.nih.gov/articles/PMC10895275/
19	R19	Histone Lactylation Boosts Reparative Gene Activation Post-Myocardial Infarction. 2022.		https://pubmed.ncbi.nlm.nih.gov/36268709/
20	R20	Shim et al. c-Myc transactivation of LDH-A: implications for tumor metabolism and growth. 1997.		https://pubmed.ncbi.nlm.nih.gov/9192621/
21	R21	Structural characterization of the apo form and NADH binary complex of human lactate dehydrogenase. 2014.		https://pmc.ncbi.nlm.nih.gov/articles/PMC4014127/

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Row	reference id	source type	checked date
1	R01	Original research	2026-10-07
2	R02	Original research	2026-10-07
3	R03	Original research	2026-10-07
4	R04	Original research	2026-10-07
5	R05	Original research	2026-10-07
6	R06	Original research	2026-10-07
7	R07	Original committee recommendations	2026-10-07
8	R08	Original research	2026-10-07
9	R09	Original research	2026-10-07
10	R10	Original research	2026-10-07
11	R11	Original research	2026-10-07
12	R12	Original research	2026-10-07
13	R13	Original research; existing manuscript reference 15	2026-10-07
14	R14	Original research; existing manuscript reference 17	2026-10-07
15	R15	Original research; existing manuscript reference 18	2026-10-07
16	R16	Official curated nomenclature and function resource	2026-10-07
17	R17	Original research	2026-10-07
18	R18	Original research	2026-10-07
19	R19	Original research	2026-10-07
20	R20	Original research	2026-10-07
21	R21	Original research	2026-10-07

Table S6 C-index increments and likelihood ratio tests

Column block 1 of 4

Row	cohort	covariate	metric	n	events	paired valid	baseline apparent
1	TCGA	age	C_index	145	89	1000	0.668793659322945
2	TCGA	FAB	C_index	145	89	989	0.583691563675443

Row	cohort	covariate	metric	n	events	paired valid	baseline apparent
3	TCGA	cyto	C_index	145	89	1000	0.585639441160666
4	TCGA	FLT3	C_index	145	89	1000	0.544666845781838
5	TCGA	NPM1	C_index	145	89	1000	0.513164965072542
6	TCGA	CEBPA	C_index	145	89	1000	0.504030091348737
7	TARGET	age	C_index	1857	672	1000	0.514541631524568
8	TARGET	cyto	C_index	1857	672	1000	0.566063235518345
9	TARGET	FLT3	C_index	1857	672	1000	0.522615872961714
10	TARGET	NPM1	C_index	1857	672	1000	0.526309391923519
11	TARGET	CEBPA	C_index	1857	672	1000	0.52062090356866
12	TCGA	Clin+Gen	C_index	145	89	989	
13	TARGET	Clin+Gen	C_index	1857	672	1000	

Column block 2 of 4

Row	cohort	covariate	augmented apparent	baseline corrected	augmented corrected	corrected delta	distribution p025
1	TCGA	age	0.66886082751209	0.670719018485975	0.664285937081187	-0.0064330814047875	-0.0481581431091312
2	TCGA	FAB	0.605454056958624	0.544267520574012	0.571867888967062	0.0276003683930494	-0.0118028006306469
3	TCGA	cyto	0.602095647501343	0.576865623314367	0.585415919426597	0.0085502961122299	-0.0371472239800317
4	TCGA	FLT3	0.597528210639441	0.539004809849187	0.586967727752869	0.0479629179036818	-0.0164259025872873
5	TCGA	NPM1	0.573750671681891	0.498500641611565	0.56254816975851	0.0640475281469447	-0.00014831430601
6	TCGA	CEBPA	0.577377753895755	0.491804548348547	0.566939198889181	0.0751346505406343	0.012349939890202
7	TARGET	age	0.586566850667318	0.509804681027479	0.58510838013322	0.0753036991057402	0.0462855515767719
8	TARGET	cyto	0.598838418141746	0.564677371001551	0.597174875569584	0.0324975045680329	0.0145161745585814
9	TARGET	FLT3	0.592340639691041	0.522529891813596	0.591055338243635	0.0685254464300389	0.0446868433711448
10	TARGET	NPM1	0.605530255614117	0.526371742130175	0.605292827667491	0.0789210855373163	0.0586754020113445
11	TARGET	CEBPA	0.596875436499516	0.520620373286961	0.596665856465924	0.0760454831789623	0.0554339156453657
12	TCGA	Clin+Gen		0.6620499448006254	0.6704847633827202	0.0084348185820948	-0.0144875755899552
13	TARGET	Clin+Gen		0.6146288871061559	0.6285490079554241	0.0139201208492681	0.0019395862691182

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Row	cohort	covariate	distribution p975	LRT	P	FDR	baseline corrected all valid
1	TCGA	age	0.0109084185800394	2.84526545249707	0.0916434568474346	0.100807802532178	
2	TCGA	FAB	0.0606835239802408	5.65427120444303	0.0174129093651664	0.0212824447796478	
3	TCGA	cyto	0.0443998430317553	1.52430763504037	0.216968781723702	0.216968781723702	
4	TCGA	FLT3	0.09622108059532	6.55648945414737	0.0104502327871156	0.0143690700822839	
5	TCGA	NPM1	0.121139237533101	7.21685432103163	0.00722221643034	0.0113491972476773	
6	TCGA	CEBPA	0.136453026759035	7.85409970783155	0.005070559526075	0.0092960257978042	
7	TARGET	age	0.102049826485279	62.2698401250382	2.9947657925576104e-15	1.64712118590669e-14	
8	TARGET	cyto	0.0496549334156712	25.3959330912094	4.66901396606745e-07	1.02718307253484e-06	
9	TARGET	FLT3	0.0902880905663369	60.6226965555634	6.913231608259211e-15	2.53485158969504e-14	
10	TARGET	NPM1	0.0996583832760565	63.4490831399016	1.6456598238342502e-15	1.64712118590669e-14	
11	TARGET	CEBPA	0.0980802198557192	55.9557520221369	7.41205607330295e-14	2.03831542015831e-13	
12	TCGA	Clin+Gen	0.0194725032793018		0.124468133832522	0.124468133832522	0.662049944800625
13	TARGET	Clin+Gen	0.0230564504953279		2.51862966210307e-05	5.03725932420614e-05	0.614628887106156

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Row	cohort	covariate	full corrected all valid	baseline valid	full valid
1	TCGA	age			
2	TCGA	FAB			
3	TCGA	cyto			
4	TCGA	FLT3			
5	TCGA	NPM1			
6	TCGA	CEBPA			
7	TARGET	age			
8	TARGET	cyto			

Row	cohort	covariate	full corrected all valid	baseline valid	full valid
9	TARGET	FLT3			
10	TARGET	NPM1			
11	TARGET	CEBPA			
12	TCGA	Clin+Gen	0.67048476338272	989.0	989.0
13	TARGET	Clin+Gen	0.628549007955424	1000.0	1000.0

Table S7 Time-dependent AUC increments

Column block 1 of 5

Row	cohort	covariate	metric	n	events	paired valid	baseline apparent
1	TCGA	age	AUC_1yr	145	89	1000	0.669872736355971
2	TCGA	age	AUC_3yr	145	89	1000	0.734100282679447
3	TCGA	age	AUC_5yr	145	89	999	0.807730782200652
4	TCGA	FAB	AUC_1yr	145	89	989	0.632565008129341
5	TCGA	FAB	AUC_3yr	145	89	989	0.666579656673386
6	TCGA	FAB	AUC_5yr	145	89	988	0.656904601809865
7	TCGA	cyto	AUC_1yr	145	89	1000	0.57501599222366
8	TCGA	cyto	AUC_3yr	145	89	1000	0.639317493762684
9	TCGA	cyto	AUC_5yr	145	89	999	0.728190499160651
10	TCGA	FLT3	AUC_1yr	145	89	1000	0.612434237613956
11	TCGA	FLT3	AUC_3yr	145	89	1000	0.564058367358959
12	TCGA	FLT3	AUC_5yr	145	89	999	0.646716232639566
13	TCGA	NPM1	AUC_1yr	145	89	1000	0.52864325872725
14	TCGA	NPM1	AUC_3yr	145	89	1000	0.521739275481973
15	TCGA	NPM1	AUC_5yr	145	89	999	0.592299408736927
16	TCGA	CEBPA	AUC_1yr	145	89	1000	0.496489943837284
17	TCGA	CEBPA	AUC_3yr	145	89	1000	0.507119745586402
18	TCGA	CEBPA	AUC_5yr	145	89	999	0.474308881174672

Row	cohort	covariate	metric	n	events	paired valid	baseline apparent
19	TARGET	age	AUC_1yr	1857	672	1000	0.51771109901075
20	TARGET	age	AUC_3yr	1857	672	1000	0.513561333682596
21	TARGET	age	AUC_5yr	1857	672	1000	0.497189941727208
22	TARGET	cyto	AUC_1yr	1857	672	1000	0.584228413384109
23	TARGET	cyto	AUC_3yr	1857	672	1000	0.575233270780497
24	TARGET	cyto	AUC_5yr	1857	672	1000	0.601471176630264
25	TARGET	FLT3	AUC_1yr	1857	672	1000	0.513949435970208
26	TARGET	FLT3	AUC_3yr	1857	672	1000	0.528944672407118
27	TARGET	FLT3	AUC_5yr	1857	672	1000	0.53766268123908
28	TARGET	NPM1	AUC_1yr	1857	672	1000	0.520503340640036
29	TARGET	NPM1	AUC_3yr	1857	672	1000	0.53485223092145
30	TARGET	NPM1	AUC_5yr	1857	672	1000	0.540029550395351
31	TARGET	CEBPA	AUC_1yr	1857	672	1000	0.520042505660731
32	TARGET	CEBPA	AUC_3yr	1857	672	1000	0.529235742772037
33	TARGET	CEBPA	AUC_5yr	1857	672	1000	0.53208920043735
34	TCGA	Clin+Gen	AUC_1yr	145	89	989	
35	TCGA	Clin+Gen	AUC_3yr	145	89	989	
36	TCGA	Clin+Gen	AUC_5yr	145	89	988	
37	TARGET	Clin+Gen	AUC_1yr	1857	672	1000	
38	TARGET	Clin+Gen	AUC_3yr	1857	672	1000	
39	TARGET	Clin+Gen	AUC_5yr	1857	672	1000	

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Row	cohort	covariate	metric	augmented apparent	baseline corrected	augmented corrected	corrected delta
1	TCGA	age	AUC_1yr	0.674761572680438	0.672408561598997	0.668847881786665	-0.0035606798123316
2	TCGA	age	AUC_3yr	0.751642959371386	0.736844046319172	0.746390226583584	0.0095461802644119
3	TCGA	age	AUC_5yr	0.828774139811132	0.80760958827535	0.817113863474155	0.0095042751988048

Row	cohort	covariate	metric	augmented apparent	baseline corrected	augmented corrected	corrected delta
4	TCGA	FAB	AUC_1yr	0.639917359755329	0.584930554963137	0.599473538182939	0.0145429832198017
5	TCGA	FAB	AUC_3yr	0.717531412278786	0.606269203767446	0.666253806440081	0.0599846026726352
6	TCGA	FAB	AUC_5yr	0.729852537918472	0.58250307607124	0.664873624274774	0.0823705482035345
7	TCGA	cyto	AUC_1yr	0.610070055470908	0.564720429988749	0.589642356883898	0.024921926895149
8	TCGA	cyto	AUC_3yr	0.698138312853299	0.625762803643802	0.674229453079779	0.0484666494359778
9	TCGA	cyto	AUC_5yr	0.739237287691827	0.72409006056533	0.718535191433313	-0.0055548691320169
10	TCGA	FLT3	AUC_1yr	0.669185821288928	0.605093109592814	0.654509785267906	0.0494166756750926
11	TCGA	FLT3	AUC_3yr	0.692008306398611	0.555470566318519	0.677747876031234	0.122277309712715
12	TCGA	FLT3	AUC_5yr	0.745115288854743	0.643515680874103	0.736719780320027	0.0932040994459234
13	TCGA	NPM1	AUC_1yr	0.593950969661916	0.51043956540863	0.579081712679142	0.0686421472705123
14	TCGA	NPM1	AUC_3yr	0.67406605758034	0.495723657001304	0.65754991673071	0.161826259729406
15	TCGA	NPM1	AUC_5yr	0.683014973718517	0.572638830115511	0.669358627619027	0.0967197975035156
16	TCGA	CEBPA	AUC_1yr	0.599145490476589	0.482420059602192	0.585722942456855	0.103302882854663
17	TCGA	CEBPA	AUC_3yr	0.679194569179788	0.487712516527896	0.664474004841452	0.176761488313557
18	TCGA	CEBPA	AUC_5yr	0.668141743277318	0.444238813696857	0.644179725417735	0.199940911720878
19	TARGET	age	AUC_1yr	0.597592214900611	0.512590063840869	0.595761544151421	0.0831714803105519
20	TARGET	age	AUC_3yr	0.6188877012846	0.507700227795038	0.617214012422343	0.109513784627305
21	TARGET	age	AUC_5yr	0.626155840846574	0.49124767887638	0.624567622403153	0.133319943526774
22	TARGET	cyto	AUC_1yr	0.615311475092925	0.58268961390687	0.613165696472796	0.0304760825659257
23	TARGET	cyto	AUC_3yr	0.630416463646957	0.573488827741602	0.628549556249981	0.0550607285083782
24	TARGET	cyto	AUC_5yr	0.647085254063536	0.600261474438898	0.645367898131039	0.0451064236921405
25	TARGET	FLT3	AUC_1yr	0.596755803632788	0.513764050165999	0.595173931561697	0.0814098813956975
26	TARGET	FLT3	AUC_3yr	0.624893101466434	0.528855802752675	0.623467097199371	0.0946112944466963
27	TARGET	FLT3	AUC_5yr	0.635548635083111	0.537486285210139	0.634082502769228	0.0965962175590882
28	TARGET	NPM1	AUC_1yr	0.609062195650646	0.520566508607886	0.608517211237855	0.0879507026299689
29	TARGET	NPM1	AUC_3yr	0.643146966056904	0.534909276513531	0.642962248944956	0.108052972431425
30	TARGET	NPM1	AUC_5yr	0.652489560932004	0.54011993298766	0.652370851807836	0.112250918820175

Row	cohort	covariate	metric	augmented apparent	baseline corrected	augmented corrected	corrected delta
31	TARGET	CEBPA	AUC_1yr	0.605792742057949	0.520079469087122	0.605324652796291	0.0852451837091688
32	TARGET	CEBPA	AUC_3yr	0.633505233137314	0.529300057457091	0.633470305894834	0.104170248437742
33	TARGET	CEBPA	AUC_5yr	0.64362535922532	0.532405754402265	0.643805928740633	0.111400174338368
34	TCGA	Clin+Gen	AUC_1yr		0.7069733064152646	0.7171714770985108	0.0101981706832461
35	TCGA	Clin+Gen	AUC_3yr		0.7424913334632748	0.7595659782990232	0.0170746448357485
36	TCGA	Clin+Gen	AUC_5yr		0.8558823335217234	0.8501173384293641	-0.0057649950923593
37	TARGET	Clin+Gen	AUC_1yr		0.6102959738224246	0.6273410058133777	0.0170450319909531
38	TARGET	Clin+Gen	AUC_3yr		0.6543175371019545	0.6753316957029736	0.021014158601019
39	TARGET	Clin+Gen	AUC_5yr		0.6937090693280413	0.7044639266415783	0.0107548573135369

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Row	cohort	covariate	metric	distribution p025	distribution p975	LRT	P
1	TCGA	age	AUC_1yr	-0.0590866752800509	0.0250381864827684	2.84526545249707	0.0916434568474346
2	TCGA	age	AUC_3yr	-0.0486126146442323	0.0393255842277877	2.84526545249707	0.0916434568474346
3	TCGA	age	AUC_5yr	-0.0933942412967909	0.0695631785678088	2.84526545249707	0.0916434568474346
4	TCGA	FAB	AUC_1yr	-0.0509836792486333	0.0756698942485027	5.65427120444303	0.0174129093651664
5	TCGA	FAB	AUC_3yr	-0.0133203807086757	0.119050907268464	5.65427120444303	0.0174129093651664
6	TCGA	FAB	AUC_5yr	-0.057094294594278	0.199960105010808	5.65427120444303	0.0174129093651664
7	TCGA	cyto	AUC_1yr	-0.0484952837295965	0.0888819678159627	1.52430763504037	0.216968781723702
8	TCGA	cyto	AUC_3yr	-0.0261387610248533	0.116252497280536	1.52430763504037	0.216968781723702
9	TCGA	cyto	AUC_5yr	-0.132878427708331	0.134487075746782	1.52430763504037	0.216968781723702
10	TCGA	FLT3	AUC_1yr	-0.048597796612078	0.119556498617218	6.55648945414737	0.0104502327871156
11	TCGA	FLT3	AUC_3yr	0.0187870930982594	0.216682036223992	6.55648945414737	0.0104502327871156
12	TCGA	FLT3	AUC_5yr	-0.0810838097435587	0.290052923650584	6.55648945414737	0.0104502327871156
13	TCGA	NPM1	AUC_1yr	-0.0298190123509505	0.164887740853232	7.21685432103163	0.00722221643034
14	TCGA	NPM1	AUC_3yr	0.0398725240126594	0.27290756038985	7.21685432103163	0.00722221643034
15	TCGA	NPM1	AUC_5yr	-0.124420325505068	0.307771047549315	7.21685432103163	0.00722221643034

Row	cohort	covariate	metric	distribution p025	distribution p975	LRT	P
16	TCGA	CEBPA	AUC_1yr	0.0106504704503375	0.196972603377178	7.85409970783155	0.005070559526075
17	TCGA	CEBPA	AUC_3yr	0.0726187625653973	0.283218518099855	7.85409970783155	0.005070559526075
18	TCGA	CEBPA	AUC_5yr	-0.0119693462067821	0.446967237184419	7.85409970783155	0.005070559526075
19	TARGET	age	AUC_1yr	0.0360742628608331	0.130139898721627	62.2698401250382	2.9947657925576104e-15
20	TARGET	age	AUC_3yr	0.0736343326850847	0.144938378475547	62.2698401250382	2.9947657925576104e-15
21	TARGET	age	AUC_5yr	0.0927991170113051	0.173163950894686	62.2698401250382	2.9947657925576104e-15
22	TARGET	cyto	AUC_1yr	0.0015603561233678	0.0564966891306337	25.3959330912094	4.66901396606745e-07
23	TARGET	cyto	AUC_3yr	0.0314590032680803	0.0775442237475321	25.3959330912094	4.66901396606745e-07
24	TARGET	cyto	AUC_5yr	0.0195046360078527	0.0690846098132382	25.3959330912094	4.66901396606745e-07
25	TARGET	FLT3	AUC_1yr	0.0460725563300122	0.112897342527162	60.6226965555634	6.913231608259211e-15
26	TARGET	FLT3	AUC_3yr	0.0638684010570659	0.121436355772447	60.6226965555634	6.913231608259211e-15
27	TARGET	FLT3	AUC_5yr	0.0631290851986257	0.128564496140025	60.6226965555634	6.913231608259211e-15
28	TARGET	NPM1	AUC_1yr	0.0564426303843044	0.117735692917004	63.4490831399016	1.6456598238342502e-15
29	TARGET	NPM1	AUC_3yr	0.0817833803670543	0.134662178298064	63.4490831399016	1.6456598238342502e-15
30	TARGET	NPM1	AUC_5yr	0.0847960891458006	0.142345765739789	63.4490831399016	1.6456598238342502e-15
31	TARGET	CEBPA	AUC_1yr	0.0540124689798818	0.116381788969632	55.9557520221369	7.41205607330295e-14
32	TARGET	CEBPA	AUC_3yr	0.0765247195600508	0.131545082121131	55.9557520221369	7.41205607330295e-14
33	TARGET	CEBPA	AUC_5yr	0.0830571305103345	0.142332935393939	55.9557520221369	7.41205607330295e-14
34	TCGA	Clin+Gen	AUC_1yr	-0.0278918844731992	0.0303150599595829		0.124468133832522
35	TCGA	Clin+Gen	AUC_3yr	-0.009011078177048	0.0379327495048909		0.124468133832522
36	TCGA	Clin+Gen	AUC_5yr	-0.0479415110959297	0.0306963219952755		0.124468133832522
37	TARGET	Clin+Gen	AUC_1yr	-0.0027247846250231	0.0332651572043417		2.51862966210307e-05
38	TARGET	Clin+Gen	AUC_3yr	0.0046941464555672	0.0332226501246021		2.51862966210307e-05
39	TARGET	Clin+Gen	AUC_5yr	-0.0069376360000315	0.0237323143712114		2.51862966210307e-05

Column block 4 of 5

Row	cohort	covariate	metric	FDR	baseline corrected all valid	full corrected all valid	baseline valid
1	TCGA	age	AUC_1yr	0.100807802532178			
2	TCGA	age	AUC_3yr	0.100807802532178			
3	TCGA	age	AUC_5yr	0.100807802532178			
4	TCGA	FAB	AUC_1yr	0.0212824447796478			
5	TCGA	FAB	AUC_3yr	0.0212824447796478			
6	TCGA	FAB	AUC_5yr	0.0212824447796478			
7	TCGA	cyto	AUC_1yr	0.216968781723702			
8	TCGA	cyto	AUC_3yr	0.216968781723702			
9	TCGA	cyto	AUC_5yr	0.216968781723702			
10	TCGA	FLT3	AUC_1yr	0.0143690700822839			
11	TCGA	FLT3	AUC_3yr	0.0143690700822839			
12	TCGA	FLT3	AUC_5yr	0.0143690700822839			
13	TCGA	NPM1	AUC_1yr	0.0113491972476773			
14	TCGA	NPM1	AUC_3yr	0.0113491972476773			
15	TCGA	NPM1	AUC_5yr	0.0113491972476773			
16	TCGA	CEBPA	AUC_1yr	0.0092960257978042			
17	TCGA	CEBPA	AUC_3yr	0.0092960257978042			
18	TCGA	CEBPA	AUC_5yr	0.0092960257978042			
19	TARGET	age	AUC_1yr	1.64712118590669e-14			
20	TARGET	age	AUC_3yr	1.64712118590669e-14			
21	TARGET	age	AUC_5yr	1.64712118590669e-14			
22	TARGET	cyto	AUC_1yr	1.02718307253484e-06			
23	TARGET	cyto	AUC_3yr	1.02718307253484e-06			

Row	cohort	covariate	metric	FDR	baseline corrected all valid	full corrected all valid	baseline valid
24	TARGET	cyto	AUC_5yr	1.02718307253484e-06			
25	TARGET	FLT3	AUC_1yr	2.53485158969504e-14			
26	TARGET	FLT3	AUC_3yr	2.53485158969504e-14			
27	TARGET	FLT3	AUC_5yr	2.53485158969504e-14			
28	TARGET	NPM1	AUC_1yr	1.64712118590669e-14			
29	TARGET	NPM1	AUC_3yr	1.64712118590669e-14			
30	TARGET	NPM1	AUC_5yr	1.64712118590669e-14			
31	TARGET	CEBPA	AUC_1yr	2.03831542015831e-13			
32	TARGET	CEBPA	AUC_3yr	2.03831542015831e-13			
33	TARGET	CEBPA	AUC_5yr	2.03831542015831e-13			
34	TCGA	Clin+Gen	AUC_1yr	0.124468133832522	0.706973306415264	0.71717147709851	989.0
35	TCGA	Clin+Gen	AUC_3yr	0.124468133832522	0.742491333463275	0.759565978299023	989.0
36	TCGA	Clin+Gen	AUC_5yr	0.124468133832522	0.855882333521724	0.850117338429364	988.0
37	TARGET	Clin+Gen	AUC_1yr	5.03725932420614e-05	0.610295973822424	0.627341005813377	1000.0
38	TARGET	Clin+Gen	AUC_3yr	5.03725932420614e-05	0.654317537101955	0.675331695702974	1000.0
39	TARGET	Clin+Gen	AUC_5yr	5.03725932420614e-05	0.693709069328041	0.704463926641578	1000.0

Column block 5 of 5

Row	cohort	covariate	metric	full valid
1	TCGA	age	AUC_1yr	
2	TCGA	age	AUC_3yr	
3	TCGA	age	AUC_5yr	
4	TCGA	FAB	AUC_1yr	
5	TCGA	FAB	AUC_3yr	
6	TCGA	FAB	AUC_5yr	
7	TCGA	cyto	AUC_1yr	

Row	cohort	covariate	metric	full valid
8	TCGA	cyto	AUC_3yr	
9	TCGA	cyto	AUC_5yr	
10	TCGA	FLT3	AUC_1yr	
11	TCGA	FLT3	AUC_3yr	
12	TCGA	FLT3	AUC_5yr	
13	TCGA	NPM1	AUC_1yr	
14	TCGA	NPM1	AUC_3yr	
15	TCGA	NPM1	AUC_5yr	
16	TCGA	CEBPA	AUC_1yr	
17	TCGA	CEBPA	AUC_3yr	
18	TCGA	CEBPA	AUC_5yr	
19	TARGET	age	AUC_1yr	
20	TARGET	age	AUC_3yr	
21	TARGET	age	AUC_5yr	
22	TARGET	cyto	AUC_1yr	
23	TARGET	cyto	AUC_3yr	
24	TARGET	cyto	AUC_5yr	
25	TARGET	FLT3	AUC_1yr	
26	TARGET	FLT3	AUC_3yr	
27	TARGET	FLT3	AUC_5yr	
28	TARGET	NPM1	AUC_1yr	
29	TARGET	NPM1	AUC_3yr	
30	TARGET	NPM1	AUC_5yr	
31	TARGET	CEBPA	AUC_1yr	

Row	cohort	covariate	metric	full valid
32	TARGET	CEBPA	AUC_3yr	
33	TARGET	CEBPA	AUC_5yr	
34	TCGA	Clin+Gen	AUC_1yr	989.0
35	TCGA	Clin+Gen	AUC_3yr	989.0
36	TCGA	Clin+Gen	AUC_5yr	988.0
37	TARGET	Clin+Gen	AUC_1yr	1000.0
38	TARGET	Clin+Gen	AUC_3yr	1000.0
39	TARGET	Clin+Gen	AUC_5yr	1000.0

Table S8 DCA comparisons with default strategies

Column block 1 of 4

Row	cohort	comparison	high model	low model	time years	threshold min	threshold max
1	TCGA	LCDI_vs_All	A	All	3	0.05	0.75
2	TCGA	LCDI_vs_None	A		3	0.05	0.75
3	TARGET	LCDI_vs_All	A	All	3	0.05	0.75
4	TARGET	LCDI_vs_None	A		3	0.05	0.75

Column block 2 of 4

Row	cohort	comparison	threshold n	mean delta NB	area delta NB	bootstrap mean	bootstrap se
1	TCGA	LCDI_vs_All	71	0.0416406658890754	0.0291484661223528	0.0459344232523017	0.0268747683564431
2	TCGA	LCDI_vs_None	71	0.4322740306887332	0.3025918214821132	0.4346660892311282	0.0667546521345652
3	TARGET	LCDI_vs_All	71	0.3213848292163532	0.2249693804514472	0.3216686045524406	0.0153131016215028
4	TARGET	LCDI_vs_None	71	0.0876202039647523	0.0613341427753266	0.0880442409658375	0.0072633906114318

Column block 3 of 4

Row	cohort	comparison	centered p025	centered p975	percentile p025	percentile p975	P bootstrap centered
1	TCGA	LCDI_vs_All	-0.0139096460070446	0.0971909777851955	0.0057531334854003	0.1080792215660413	0.1041372351160444
2	TCGA	LCDI_vs_None	0.3019902779395067	0.5625577834379597	0.3076626845058651	0.5706717231117561	0.0002018163471241

Row	cohort	comparison	centered p025	centered p975	percentile p025	percentile p975	P bootstrap centered
3	TARGET	LCDI_vs_All	0.2912363540024559	0.3515333044302505	0.2920961657222548	0.3520752719545199	0.0001999600079984
4	TARGET	LCDI_vs_None	0.0734910178150419	0.1017493901144627	0.0742112522470812	0.1025674768224708	0.0001999600079984

Column block 4 of 4

Row	cohort	comparison	P normal approx	bootstrap valid	optimism corrected mean delta NB	FDR	FDR original 10 tests
1	TCGA	LCDI_vs_All	0.1212775504413355	4954	0.0343239589463015	0.1041372351160444	0.1041372351160444
2	TCGA	LCDI_vs_None	9.445835905367123e-11	4954	0.4249573237459594	0.00026908846283213336	0.0002883090673201
3	TARGET	LCDI_vs_All	8.518683270982488e-98	5000	0.3207182799894221	0.00026908846283213336	0.0002883090673201
4	TARGET	LCDI_vs_None	1.6510575434720606e-33	5000	0.0869536547378212	0.00026908846283213336	0.0002883090673201

Table S9 qPCR primers

Row	primer	sequence 5to3	product bp
1	H-MPO-F	ACCCGTTCCAGTGAGATGC	274
2	H-MPO-R	TGAAGACGTTGGCGATGC	
3	H-CREB1-F	AATCTGGAGCCGAGAACC	269
4	H-CREB1-R	GAGTTACGGTGGGAGCAG	
5	H-CASP1-F	GACAAGGGTGCTGAACAA	96
6	H-CASP1-R	GGAATAACGGAGTCAATCA	
7	H-SLC16A4-F	TACCAAGTGGCTGCTGTG	148
8	H-SLC16A4-R	GGGCTCCTGTCCAGTCATAC	
9	H-GPX4-F	CCTTTGCCGCCTACTGAA	144
10	H-GPX4-R	ATGTGCCCGTCGATGTCC	
11	H-GAPDH-F	TGACTTCAACAGCGACACCCA	121
12	H-GAPDH-R	CACCCTGTTGCTGTAGCCAAA	
13	H-ACTB-F	GCGTGACATTAAGGAGAAGC	236
14	H-ACTB-R	CCACGTCACACTTCATGATGG	

Table S10 State and signature gene sets

Row	module	gene
1	GSE130756_Mono	CST3
2	GSE130756_Mono	FCER1G
3	GSE130756_Mono	S100A11
4	GSE130756_Mono	LYZ
5	GSE130756_Mono	CTSS
6	GSE130756_Mono	S100A8
7	GSE130756_Mono	S100A9
8	GSE130756_Mono	TYROBP
9	GSE130756_Mono	MS4A6A
10	GSE130756_Mono	FCN1
11	GSE130756_Mono	LGALS1
12	GSE130756_Mono	PSAP
13	GSE130756_Mono	ANXA2
14	GSE130756_Mono	CTSD
15	GSE130756_Mono	LST1
16	GSE130756_Mono	LGALS3
17	GSE130756_Prog	SOX4
18	GSE130756_Prog	CD99
19	GSE130756_Prog	CFD
20	GSE130756_Prog	FOS
21	GSE130756_Prog	ITM2A
22	GSE130756_Prog	TMSB10
23	GSE130756_Prog	SH3BGRL3
24	GSE130756_Prog	MGST1
25	GSE130756_Prog	MRPL33
26	GSE130756_Prog	MARCKSL1

Row	module	gene
27	GSE130756_Prog	GATA2
28	GSE130756_Prog	HOPX
29	GSE130756_Prog	CD34
30	GSE130756_Prog	SPINK2
31	Neutrophil	CSF3R
32	Neutrophil	FCGR3B
33	Neutrophil	FFAR2
34	Neutrophil	CXCR1
35	Neutrophil	CXCR2
36	Neutrophil	MPO
37	Neutrophil	ELANE
38	Neutrophil	CTSG
39	Endpoint_progenitor	CD34
40	Endpoint_progenitor	AVP
41	Endpoint_progenitor	CRHBP
42	Endpoint_progenitor	MEIS1
43	Endpoint_progenitor	HOPX
44	Endpoint_progenitor	PROM1
45	Endpoint_progenitor	MLLT3
46	Endpoint_progenitor	HLF
47	Endpoint_progenitor	KIT
48	Endpoint_progenitor	GATA2
49	Endpoint_granulocytic	CSF3R
50	Endpoint_granulocytic	FCGR3B
51	Endpoint_granulocytic	CEACAM8
52	Endpoint_granulocytic	S100A8
53	Endpoint_granulocytic	S100A9

Row	module	gene
54	Endpoint_granulocytic	ELANE
55	Endpoint_granulocytic	CTSG
56	Endpoint_granulocytic	AZU1
57	Endpoint_granulocytic	LTF
58	Endpoint_monocytic	LST1
59	Endpoint_monocytic	FCN1
60	Endpoint_monocytic	CTSD
61	Endpoint_monocytic	LILRB1
62	Endpoint_monocytic	CTSS
63	Endpoint_monocytic	FCGR3A
64	Endpoint_dendritic	FCER1A
65	Endpoint_dendritic	CLEC10A
66	Endpoint_dendritic	CD1C
67	Endpoint_dendritic	GZMB
68	Endpoint_dendritic	JCHAIN